Preserving Lesion Quantification in Longitudinal Whole-Body PSMA PET/CT Registration

Fryderyk Kögl[a,b,c,*], **Anna Reithmeir**[a,b,c], **Daniel M. Lang**[a], **Emily M. Chan**[a,b], **Lukas Förner**[d], **Rickmer Braren**[e], **Julia A. Schnabel**[a,b,c,f]

[a]Institute of Machine Learning in Biomedical Imaging, Computational Health Center, Helmholtz Munich, Ingolstädter Landstraße 1, 85764 Neuherberg, Germany

[b]Computational Imaging and AI in Medicine, TUM School of Computation, Information and Technology, Technical University of Munich, Lichtenbergstraße 2a, 85748 Garching, Germany

[c]Munich Center for Machine Learning, Mottlstraße 1, 80804 Munich, Germany

[d]Department of Diagnostic and Interventional Radiology and Neuroradiology, University Hospital Augsburg, Stenglinstraße 2, 86156 Augsburg, Germany

[e]Department of Diagnostic and Interventional Radiology and Nuclear Medicine, University Medical Center Hamburg-Eppendorf, Martinistraße 52, 20246 Hamburg, Germany

[f]Department of Biomedical Computing, School of Biomedical Engineering & Imaging Sciences, King's College London, St Thomas' Hospital, Westminster Bridge Road, London SE1 7EH, United Kingdom

Abstract. **Purpose:** Registering follow-up to baseline PSMA PET/CT images enables voxel-wise comparison of tumor burden, but the deformation can itself change the metabolic tumor volume (MTV) and total lesion glycolysis (TLG) of lesions. We examine how PET-preservation and bone-rigidity losses trade alignment for less registration-induced change, and whether they belong in network training or instance optimization (IO).

Approach: We trained LapIRN networks with and without the losses on the Learn2Reg 2026 PSMAReg data and applied the losses during training, IO, or both. We evaluated three rigidity penalties and compared the resulting method with tuned NiftyReg and ConvexAdam on 28 held-out patients using patient-level statistics and per-lesion analyses.

Results: With the losses only in IO, our method reached 78.3 % CT Dice (NiftyReg 71.8 %, ConvexAdam 60.3 %); no difference in MTV or TLG change from either baseline was detected. Without the losses, the mean MTV and TLG changes rose after IO relative to network output; with the losses, IO gained 2.9 Dice points without a detected difference in either image-level measure. Applying the losses only in IO gained 1.0 Dice point over applying them in both training and IO; whether preservation differs between these placements remains unresolved. Lesions missed by the predicted masks contributed more than half of the residual image-level change in the PET-loss IO analysis. With provided masks, the image-level MTV change fell to 0.1 %, but individual lesions still changed by a median of 13.5 %. A per-bone rigid fit produced the lowest within-bone distance distortion among the tested penalties (13.4 % to 5.0 %). When added to the PET losses, it produced no detected further reduction in lesion change.

Conclusions: With predicted lesion masks, IO with PET-preservation and bone-rigidity losses improved anatomical alignment without a detected difference in image-level MTV or TLG change relative to network output. This option requires time for per-pair refinement; when IO was omitted, training with the losses lowered the mean MTV and TLG changes, with a detected reduction only for MTV. Missed lesions limit image-level preservation, while substantial changes to individual lesions remain even with provided masks. Per-lesion evaluation is therefore needed to judge preservation for lesion-level applications.

Keywords: deformable image registration, longitudinal imaging, biomarker preservation, instance optimization.

***Corresponding author:** Fryderyk Kögl, fryderyk.koegl@helmholtz-munich.de

1 Introduction

Prostate-specific membrane antigen (PSMA) PET/CT is now routinely used for staging prostate cancer, for planning and following radioligand therapy, and for assessing treatment response.[1] In such longitudinal imaging, patients are scanned in a baseline session before therapy and in one or more follow-up sessions. Registering each follow-up image to the baseline image makes it possible

to compare tumor burden voxel by voxel or track individual lesions over time.[2] For these uses, a registration has to align the anatomy while preserving the PET-derived quantities of the warped image. Two such quantities, under the names used by the Learn2Reg 2026 PSMAReg challenge,[2] are the metabolic tumor volume (MTV) and the total lesion glycolysis (TLG); for PSMA PET, they measure the PSMA-positive tumor volume and the total lesion PSMA uptake. A deformation that compresses or stretches a lesion would change both these quantities and thereby introduce a registration-induced bias into the measured longitudinal change. When the registered follow-up is used to match lesions or compare them with baseline, such processing-induced changes could be mistaken for part of the longitudinal change and could therefore affect lesion tracking or response assessment. The challenge, part of the Learn2Reg series[3,4] and based on the PSMA-PET-CT-Lesions dataset,[5] formalizes this setting by registering each follow-up image to its baseline image with a dense deformation field. It scores organ overlap, boundary distance, folding of the deformation field, and the change of MTV and TLG caused by the deformations. Anatomical alignment and the preservation of these quantities can compete: limiting deformation may protect lesion measurements while leaving the anatomy misaligned, whereas improving alignment may distort the lesions being measured.

Classical registration solves an optimization problem per image pair;[6–8] this can be accurate,[3,4] but it is often computationally expensive, in particular for large three-dimensional whole-body images.[9] Learning-based registration amortizes this optimization: a network is trained once on many image pairs and then registers a new pair in a single forward pass.[10–12] Instance optimization (IO) refines the network's deformation field for each image pair by classical iterative optimization at test time. Whole-body images of the same patient can differ by large deformations, for example through different positioning of the arms, spine, and bowel. We therefore build on LapIRN,[13] a learning-based method that estimates the deformation coarse to fine on a Laplacian image pyramid. Its coarse levels recover large deformations and its finer levels add smaller ones, while the transformation remains smooth and invertible. With a network and optional IO, constraints that preserve MTV and TLG can act during training, during per-pair refinement, or at both stages. IO generally improves registration accuracy but adds runtime for every image pair.[4] The label-guided IO studied here uses predicted organ and lesion masks for each pair; we evaluate its output against the provided labels. In IO, the PET-preservation losses act on the predicted lesion mask. Lesions missing from this mask are therefore not directly constrained.

In previous work, we developed a registration pipeline for the PSMAReg challenge that combines an affine pre-registration, LapIRN, and IO with PET-preservation and per-bone rigidity losses.[14] There, selected configurations were evaluated on the challenge validation cases using surrogate labels. In this extension, we retrain on a new patient-level split of the publicly released, labeled training data of the PSMAReg challenge[2,5] and reserve 28 patients for final evaluation. The 28 patients form a held-out internal test set for the present analyses, not an independent external cohort.

Our research questions are:

1. How does applying PET-preservation and bone-rigidity losses during network training, IO, or both affect the trade-off between anatomical alignment and registration-induced MTV and TLG change?

2. What limits loss-guided IO's preservation of MTV and TLG, as revealed by predicted-mask coverage and changes to individual lesions?

3. Can bone-rigidity penalties support PET preservation while maintaining the internal geometry of bones in low-resolution whole-body CT?

We answer these questions on a held-out patient split using patient-level statistics, per-lesion analyses, and comparisons with tuned registration baselines. The main contributions are:

- We systematically compare applying the PET-preservation and bone-rigidity losses during network training only, during IO only, or during both stages. Applying the losses only during IO gives higher alignment accuracy than applying them during both stages. Relative to network output, IO with the PET-preservation and bone-rigidity losses improves alignment without a detected increase in image-level MTV or TLG change.

- A mask-coverage and per-lesion analysis identifies two limits: missed lesions dominate the residual image-level error, while image totals conceal substantial changes to individual lesions even when provided masks are used.

- We compare three bone-rigidity penalties: the local Staring penalty, the same penalty applied to eroded bone masks, and a per-bone rigid-fit penalty that estimates one rigid motion for each labeled bone. The per-bone rigid fit avoids constraining tissue across bone boundaries or discarding most bone voxels through mask erosion, and it best preserves distances between points within the same bone.

2 Materials and Methods

2.1 PSMAReg Challenge Data and Study Split

We used the training set of the PSMAReg challenge, a longitudinal whole-body PSMA PET/CT cohort. Each session consists of a CT and a PET image (in SUV), resampled to $2.7344 \times 2.7344 \times 3.27$ mm and cropped to $192 \times 192 \times 288$ voxels. For every session, the challenge provides 117 CT organ labels (the TotalSegmentator[15] classes) and a PSMA lesion mask. Following the challenge, a follow-up image is always the moving image and the corresponding baseline image the fixed image. The original dataset publication reports that release of the anonymized retrospective imaging data was approved by the Ethics Committee of the Medical Faculty, LMU Munich (applications 21-1230 and 21-0098).[16] It also reports approval by the institutional data security and privacy review board. It also reports that the ethics committee waived individual informed consent because the images were fully anonymized. The present study is a secondary analysis of publicly released challenge data and involved no patient recruitment or contact.

We split the 146 labeled patients with at least two sessions at the patient level into training, validation, and test sets, aiming at 70 %/15 %/15 % of the registration pairs and balancing the sets for tumor burden and for bone- versus soft-tissue lesion location. Training uses every chronologically ordered image pair for each patient, including pairs between follow-up images. Validation and testing register each follow-up image directly to the baseline image, as in the challenge. The training/validation/test sets contained 92/26/28 patients and yielded 178/38/38 registration pairs, respectively. The split was built once with a fixed seed and frozen before any experiments of this study. All settings, checkpoints, and primary method choices were selected on the validation set. The test set was used for the final evaluation and for analyses explicitly identified as post hoc or diagnostic.

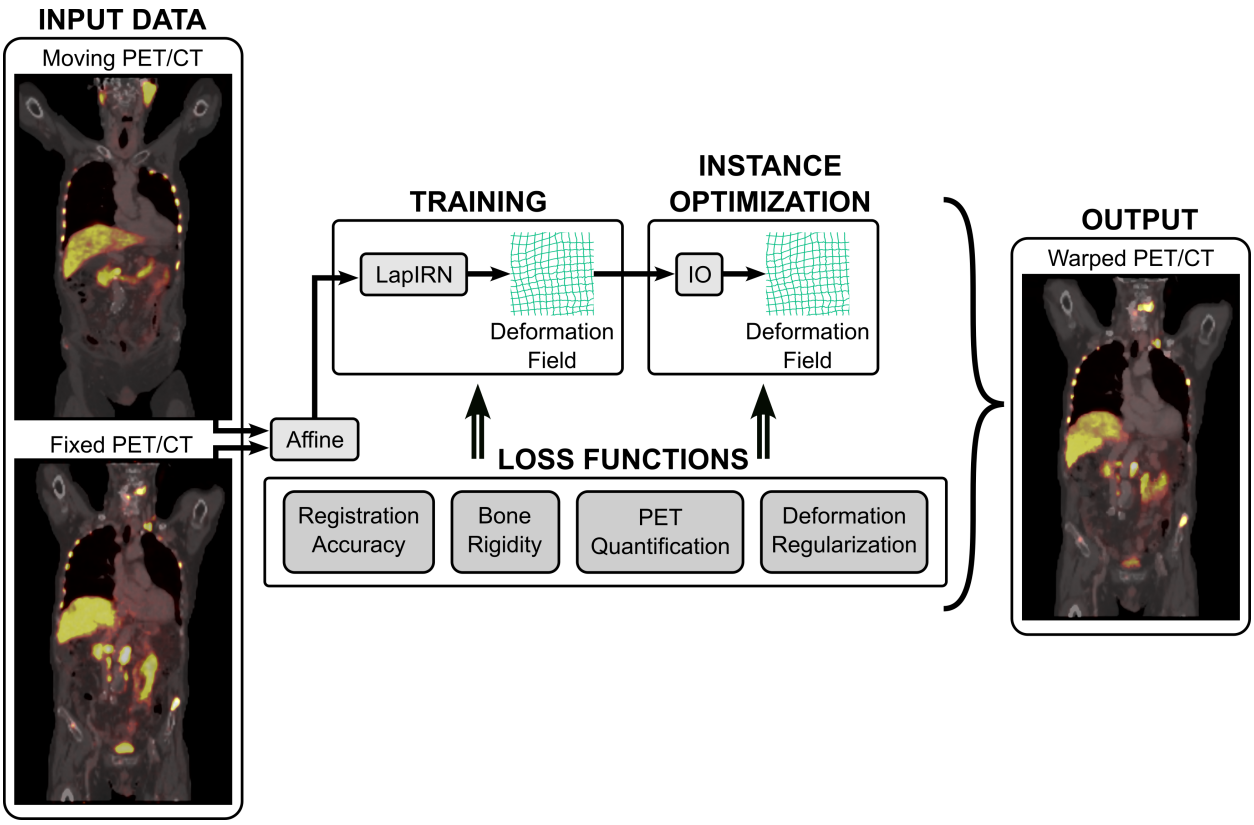


Fig 1 Our proposed registration pipeline. The follow-up (moving) and baseline (fixed) PET/CT images are aligned by an affine pre-registration, then by the LapIRN network, which predicts a dense deformation. Instance optimization (IO) refines this deformation per pair before the moving image is resampled into the fixed frame. Network training and IO use the same groups of loss terms: registration accuracy, bone rigidity, PET quantification, and deformation regularization (Sec. 2.3). The bone-rigidity and PET-quantification terms can be used in training, in IO, or in both, whereas our method uses them only in IO.

Segmentation labels in training, IO, and evaluation. Network training uses the provided organ and lesion labels of the training patients. In the main experiments, IO requires labels for each image pair but uses only predicted labels. The provided validation and test labels are reserved for evaluation (Sec. 3.1).

Organ labels are predicted from CT with TotalSegmentator.[15] Lesion masks are predicted from PET/CT with a residual-encoder nnU-Net (ResEnc-L)[17] configured according to the winning autoPET III solution.[18] The network uses 192^3 patches, the pretrained weights released by the winning team, an auxiliary organ-segmentation head, and misalignment augmentation.

The lesion model was trained by patient-grouped five-fold cross-validation on all 494 images from the non-test patients, including patients with a single session. Each validation image is segmented by the fold model for which its patient was held out, preventing IO tuning from using a prediction produced by a model trained on the same patient. Each test image is segmented by an ensemble of the five fold models. The resulting lesion masks achieve mean Dice scores of 0.56 on the validation images and 0.51 on the test images, with corresponding medians of 0.69 and 0.67.

Preprocessing. For the network and IO, CT was clipped to $[-1000, 1500]$ HU and PET to $[0, 20]$ SUV, and both were rescaled to $[0, 1]$. As in the ConvexAdam baseline of the challenge,[19] the patient bed was removed from the CT. All PET quantities in the evaluation are computed on the original SUV images.

PET measures. The challenge uses MTV and TLG to measure how registration changes the moving image's lesion volume and uptake. For a PET image P in SUV, binary lesion mask M, and voxel volume v, they are

$$\mathrm{MTV}(M) = v \sum_x M(x), \qquad \mathrm{TLG}(P, M) = v \sum_x P(x) M(x). \tag{1}$$

Both quantities are computed on the same moving image before and after warping it to the fixed frame; as in the challenge scorer, both PET and the lesion mask are warped by nearest-neighbor interpolation. The reported preservation error for $Q \in \{\mathrm{MTV}, \mathrm{TLG}\}$ is its absolute relative change,

$$|\Delta Q| = \frac{|Q_{\mathrm{after}} - Q_{\mathrm{before}}|}{\max(Q_{\mathrm{before}}, 10^{-6})}. \tag{2}$$

This measures change induced by registration, not change in disease burden between sessions.

2.2 Registration Pipeline

The pipeline (Fig. 1) has three stages, all estimated from the moving image (C_m, P_m) and the fixed image (C_f, P_f), where C denotes CT and P PET. We denote the affine sampling transform by φ_{aff}, the network sampling transform by φ_θ, and their composition by φ. For each location x on the fixed grid, $\varphi(x)$ gives the location sampled in the original moving image: the warped image is $\widetilde{C}_m(x) = C_m(\varphi(x))$, and the same convention applies to PET and labels.

Affine pre-registration. ANTs[20] estimates an affine transform from the two CTs, using body masks, a $[-1000, 1000]$ HU window, half-resolution images, and Mattes mutual information. It is converted to a dense deformation field.

Deformable network. We use LapIRN[13] with three pyramid levels at quarter, half, and full resolution. Each level predicts a stationary velocity field that is added to the upsampled velocity of the coarser level and integrated by scaling and squaring[21] into φ_θ. Each level receives four channels: the fixed CT and PET and the affinely aligned moving CT and PET. The levels are trained one after the other; a finer level is initialized from the coarser one, which is first kept frozen and then trained jointly. Each sub-network has 24 base channels and a trunk of ten residual blocks with an expansion factor of two.

Instance optimization. At test time, the deformation of each pair is refined by optimizing a residual stationary velocity field, initialized at zero and integrated by scaling and squaring (seven steps), on top of the total transformation φ. The residual is optimized at full resolution with Adam[22] (learning rate 0.025, 40 iterations). The objective is the same as in training (Sec. 2.3), with the predicted labels in place of the provided labels. Of the 40 iterates, we return the one with the lowest objective after replacing the interpolated (soft) MTV and TLG terms with their nearest-neighbor (hard) values used by the evaluation. This selection uses only predicted labels.

2.3 Registration Objective

Network training and IO minimize a weighted sum of four groups of terms (Fig. 1):

$$\mathcal{L} = \underbrace{\lambda_{\mathrm{NCC}}\mathcal{L}_{\mathrm{NCC}} + \lambda_{\mathrm{DSC}}\mathcal{L}_{\mathrm{DSC}}}_{\text{registration accuracy}} + \underbrace{\lambda_{\mathrm{MTV}}\mathcal{L}_{\mathrm{MTV}} + \lambda_{\overline{J}}\mathcal{L}_{\overline{J}} + \lambda_{J}\mathcal{L}_{J} + \lambda_{\mathrm{TLG}}\mathcal{L}_{\mathrm{TLG}}}_{\text{PET quantification}} + \underbrace{\lambda_{\mathrm{rig}}\mathcal{L}_{\mathrm{rig}}}_{\text{bone rigidity}} + \underbrace{\lambda_{\mathrm{smooth}}\mathcal{L}_{\mathrm{smooth}} + \lambda_{\mathrm{NDV}}\mathcal{L}_{\mathrm{NDV}}}_{\text{deformation regularization}}. \tag{3}$$

All terms are evaluated on the total transformation φ. The PET-preservation terms and the rigidity term are the subject of this study; they can be switched on in network training, in IO, or in both (Sec. 3). The two coarse network levels never use them, because lesions and thin bones span only a few voxels at these resolutions.

Registration accuracy. $\mathcal{L}_{\mathrm{NCC}}$ is the negative local normalized cross-correlation of the CTs, summed over one to three scales depending on the pyramid level. PET is not used for similarity, because uptake changes between sessions with disease and therapy and a PET similarity would reward matching these changes. $\mathcal{L}_{\mathrm{DSC}}$ is a soft Dice loss on the CT organ labels.

PET quantification. The PET-preservation terms keep the PET measures of Eq. (1) unchanged by the registration. For the differentiable loss terms, let M be the moving lesion mask and $\widetilde{M}$, $\widetilde{P}$ the linearly warped mask and PET. Nearest-neighbor warping, as used in the evaluation, is piecewise constant in the displacement and gives no useful gradient; linear warping gives a soft

lesion boundary instead. The volume and uptake terms penalize the relative change defined above, using the soft mask,

$$\mathcal{L}_{\mathrm{MTV}} = \left(\frac{\left| \sum_x \widetilde{M}(x) - \sum_x M(x) \right|}{\sum_x M(x) + \epsilon} \right)^2, \qquad \mathcal{L}_{\mathrm{TLG}} = \frac{\left| \sum_x \widetilde{P}(x)\widetilde{M}(x) - \sum_x P_m(x)M(x) \right|}{\sum_x P_m(x)M(x) + \epsilon}, \tag{4}$$

with $\epsilon = 10^{-5}$. Two further terms act on $\det J_\varphi(x)$, the local volume scale of the total affine-plus-deformable sampling transform, computed by central differences on the fixed voxel grid. A determinant of one preserves local volume; values above and below one correspond to expansion and contraction, respectively, of the sampled region in the moving image (the opposite change in the displayed warped lesion). Let W_s and W_h be the linearly and nearest-neighbor warped moving lesion masks, respectively, on that grid. Then

$$\mathcal{L}_{\overline{J}} = \left(\frac{\sum_x W_s(x) \det J_\varphi(x)}{\sum_x W_s(x) + \epsilon} - 1 \right)^2, \qquad \mathcal{L}_J = \frac{\sum_x W_h(x) \big(\det J_\varphi(x) - 1 \big)^2}{\sum_x W_h(x) + \epsilon}. \tag{5}$$

$\mathcal{L}_{\overline{J}}$ constrains the mean volume scale over all lesions, but allows expansion in one region to cancel contraction in another; $\mathcal{L}_J$ penalizes these local changes separately. The mask W_s is held fixed when differentiating $\mathcal{L}_{\overline{J}}$, and the hard mask W_h is not differentiable with respect to displacement; both losses differentiate through the determinant.

Bone rigidity. Many PSMA lesions lie in bone, which moves rigidly between sessions. For each of the 61 skeletal labels of the fixed image (in IO: the predicted labels), we fit a rigid transformation to the displacement of the structure in closed form.[23,24] We penalize the mean squared residual of its voxels, and $\mathcal{L}_{\mathrm{rig}}$ is the mean over structures with at least 50 voxels. Each bone is fitted independently, so neighboring bones may move relative to each other, and the term involves no finite differences, so it does not constrain the tissue around a bone.

Deformation regularization. $\mathcal{L}_{\mathrm{smooth}}$ is the mean squared spatial gradient of the displacement. $\mathcal{L}_{\mathrm{NDV}}$ is a differentiable version of the challenge's non-diffeomorphic volume, which splits every voxel into tetrahedra and accumulates the negative part of their Jacobian determinants.[25]

Weights. In training, the alignment weights are the same for all models, and the PET and rigidity weights are either used or set to zero. The IO weights were tuned on the validation set (Sec. 3.3). The numerical weights are listed in Appendix 13.

2.4 Rigidity Penalties Compared

Besides the per-bone rigid fit, we evaluate the rigidity penalty of Staring et al.,[26] which requires the Jacobian J inside a bone to be a rotation: $\det J = 1$, $J^\top J = I$, and vanishing second derivatives of the deformation. It is computed with finite differences, whose stencil reads the 6 face and 12 edge neighbors of a voxel. At a bone boundary, part of the stencil lies outside the bone, so the penalty also acts on the surrounding soft tissue or on a neighboring bone. A common remedy is

to erode the bone mask so that the stencil stays inside; we erode each bone by exactly this 18-neighbor footprint, which removes the leakage completely. The three variants (Staring, Staring with erosion, per-bone rigid fit) are compared in IO (Sec. 3).

2.5 Training

The three pyramid levels are trained sequentially with Adam,[22] with each finer level initialized from and then jointly optimized with the coarser levels. Training uses intensity, left–right, and axial field-of-view augmentations designed to mimic longitudinal variability. The full schedule, numerical precision, augmentations, hardware, and runtimes are reported in Appendix 13. The two coarse levels are shared by all models; the variants of this study differ only in the losses of the full-resolution level. Checkpoints were selected on the validation set: the coarse levels by CT Dice, the full-resolution level by the best accuracy (CT Dice and HD95) among checkpoints whose validation folding stays below $1.4 \cdot 10^{-4}\,\%$ non-diffeomorphic volume. We did not select on MTV or TLG, so that the PET losses are not credited with effects of the checkpoint choice.

Use of AI tools. We used the large language model Claude Opus 5.5 (Anthropic) for two purposes: to help write parts of the code for training, evaluation, statistical analysis, and figure generation, and to edit the language of the manuscript. All generated code was reviewed and tested by the authors, and all text edits were reviewed by the authors, who take full responsibility for the content, analyses, and results of this paper.

3 Experiments

3.1 Evaluation Metrics

All metrics are computed with the provided labels of the test images by evaluating the full-resolution deformation fields directly. The moving image is warped with corner-aligned grid coordinates, and PET and labels are sampled by nearest-neighbor interpolation. Deformation fields of the baselines are converted to the same convention; ConvexAdam's fields, which its implementation computes with edge-aligned coordinates, are converted exactly before evaluation. Of the 28 test patients (38 pairs), 26 patients (30 pairs) have lesions in the moving image and contribute to the PET metrics.

Alignment. CT Dice and the 95th percentile of the Hausdorff distance (HD95), averaged over the 117 organ labels.

PET-preservation metrics. We compute $|\Delta\mathrm{MTV}|$ and $|\Delta\mathrm{TLG}|$ as defined in Eq. (2), using the provided moving-image lesion masks. "Initial" denotes the unregistered, unresampled images; their MTV and TLG change is zero by construction and does not estimate an effect of resampling.

Per-lesion change. The aggregate errors of Eq. (2) can hide large changes of individual lesions that cancel. We therefore also split the moving lesion mask into 26-connected components and compute $|\Delta\mathrm{MTV}|$ for each. The 30 pairs with lesions contain 1435 components. For the per-lesion size and location analysis, we exclude 108 components smaller than five voxels: a one-voxel difference changes their volume by at least 25 %, making their relative changes highly quantized. This leaves 1327 lesions, grouped by size (small < 0.5 ml, medium < 5 ml, large) and location (bone-associated if at least half of the lesion lies on a skeletal label). Image-level PET metrics and the lesion-coverage analysis retain all 1435 components so that the coverage decomposition includes the full evaluated lesion volume.

Deformation quality. Percentage of non-diffeomorphic volume (NDV) as defined by the challenge,[25] within the challenge's mask (voxels where the fixed CT exceeds its minimum intensity).

Bone metrics. For the rigidity comparison, we additionally compute Dice and HD95 separately over bones and soft-tissue organs. We measure within-bone distance distortion as the mean relative change in physical distance between randomly sampled pairs of points in the same fixed-image bone, including its boundary voxels. We compute this separately for short (5–15 mm) and long (≥ 15 mm) distances; pairs below 5 mm are excluded because their relative change is sensitive to interpolation and discretization at this image resolution. Short distances characterize local deformation, whereas long distances characterize deformation across a larger part of the bone. We also compute NDV inside bones and in a three-voxel band around them and, as a secondary measure, the residual of a rigid fit to each bone's displacement (bone rigidity error). We also measure the fraction of stencil reads that leave the bone of the center voxel (stencil leakage) and the fraction of bone voxels and of bones that remain after erosion for the eroded Staring penalty. Erosion does not restrict the point pairs used for distance distortion.

3.2 Statistical Analysis

A patient may contribute several pairs and many lesions. We average each metric over a patient's eligible pairs before analysis: CT Dice, HD95, and NDV comparisons include 28 test patients, whereas MTV and TLG comparisons include the 26 patients with moving-image lesions. Paired comparisons use only patients with values for both methods and a two-sided Wilcoxon signed-rank test.[27] The implementation doubles the smaller one-sided p-value, drops zero paired differences, assigns average ranks to tied nonzero magnitudes, and returns $p = 1$ when all differences are zero. Holm correction[28] is applied separately per metric within each family. The families comprise three comparisons against "None" (no losses) in each ablation block, three IO-only versus training-plus-IO comparisons, four leave-one-out comparisons, and comparisons of our method against the other baseline-table methods. The last family contains nine comparisons for CT, HD95, and NDV and eight for MTV and TLG because "Initial" has zero PET change by construction. In the rigidity comparison, the three penalties are compared with IO without rigidity, and separately with each other (three pairs), each as one family per metric. Single post-hoc comparisons (the trade-off and recall-mask runs against their reference) are reported with uncorrected p-values; per-lesion medians are descriptive. We call a difference significant if its adjusted $p < 0.05$. Confidence intervals are 95 % percentile patient-bootstrap intervals:[29] image-level plots show intervals for

method means, whereas intervals stated for differences between two methods are for paired patient-mean differences. For per-lesion medians, patients are resampled together with all their lesions (cluster bootstrap).

For the comparison with baselines, we additionally apply the challenge's significance-based composite ranking and report 95 % patient-bootstrap rank intervals. The pairwise tests, component scores, omitted metrics, aggregation formula, and bootstrap procedure are specified in Appendix 13.

3.3 Tuning and Model Selection

All tuning used the validation set. For our IO settings, the choice of our method, and the baselines, we use one selection rule: among the candidates of a grid whose CT Dice is within two points of the most accurate one, we keep the one with the lowest $|\Delta\mathrm{MTV}| + |\Delta\mathrm{TLG}|$. The two-point tolerance is a design choice that allows some alignment accuracy to be exchanged for lower PET change; it is not a clinically established threshold. We do not use the challenge ranking for this selection because it is decided by accuracy. Dice differences between candidates are consistently significant, whereas MTV and TLG differences rarely are, so the ranking almost always retains the most accurate candidate regardless of preservation. For IO, our selection rule selected a learning rate of 0.025, 40 iterations, and a PET-weight factor of $\times 0.1$ (Table 4). Neither the selection rule nor the ranking rewards rigidity, which costs Dice without reducing the PET metrics or folding, so both would set λ_{rig} to zero. We fixed $\lambda_{\mathrm{rig}} = 0.25$ so that the selected method retained the bone-rigidity constraint evaluated in this study. Its inclusion and numerical weight are therefore design choices rather than outputs of the selection rule. The full IO and baseline grids, selected baseline settings, and edge-of-grid checks are reported in Appendix 13.

Choice of our method. Before the experiments, we defined our full method as the network trained with both losses followed by IO with both losses. Instead, we select our method on the validation set among the IO variants of Sec. 3.4 and IO directly after the affine pre-registration. We use the same rule as for the IO settings: the variant with the lowest $|\Delta\mathrm{MTV}|+|\Delta\mathrm{TLG}|$ among those within two Dice points of the most accurate one. This selected the network trained without the losses followed by IO with both losses (79.2 % Dice, $|\Delta\mathrm{MTV}|$ 6.9 %, $|\Delta\mathrm{TLG}|$ 8.9 %; the most accurate variant, IO without the losses, reached 80.8 %). The variants with the losses in training and IO changed the lesions slightly less (e.g. 5.8 % and 7.9 % with both losses) but were 2.6 points less accurate than the most accurate variant. We refer to the selected variant as our method and report the predefined full method and IO without the losses alongside it in the comparison with baselines.

3.4 Comparisons

Effect of the losses. Four networks were trained with the same alignment and regularization terms and with neither, the PET-preservation, the rigidity, or both groups of losses. Each is evaluated as network output and with IO using the same losses as in its training.

Losses in training or only in IO. The network trained without the losses is combined with IO using PET preservation, rigidity, or both, and compared with the networks trained with these losses (and IO with the same losses). To measure what the network contributes, IO with both losses is also run directly after the affine pre-registration.

Comparison with baselines. Initial alignment, the affine pre-registration, NiftyReg, ConvexAdam, IO after the affine, and our variants are compared on the test set. All registrations use the same follow-up-to-baseline pairs and are scored with the same provided organ and lesion labels and original-SUV PET images. NiftyReg and ConvexAdam use CT alone for registration, with no organ or lesion labels; both remove the patient bed, but each uses its own affine step and CT preprocessing (NiftyReg: `reg_aladin` on bed-removed CT; ConvexAdam: ANTs with a $[-300, 300]$ HU window). Our method uses CT and PET, the affine pre-registration described above, provided labels in network training, and predicted labels during IO. The NiftyReg and ConvexAdam settings were selected on the validation pairs with the same selection rule as our settings (Sec. 3.3); the rule does not consider folding. Runtimes are per pair on one NVIDIA RTX A6000 GPU and include the affine step.

Accuracy–preservation trade-off. As a post-hoc test-set sensitivity analysis, IO on the network trained without the losses is run without rigidity over PET-weight factors from zero to one, with denser sampling below the selected factor. The factor $\times 0.1$ had already been selected on the validation set and is not reselected from this test-set sweep. The tested factors are listed in Appendix 13. Besides the evaluation on the provided labels, we compute CT Dice, $|\Delta\mathrm{MTV}|$, and $|\Delta\mathrm{TLG}|$ on the predicted labels that IO optimizes, with the same definitions. To probe the effect of lesion-mask quality, we additionally rerun IO with the provided lesion masks instead of the predicted ones (organ labels remain predicted), at the tuned and full PET weights. This diagnostic run uses labels unavailable in the intended test-time setting.

Lesion coverage in predicted masks. Each provided lesion component of the moving image is assigned to a group by the fraction of its voxels covered by the predicted lesion mask: missed ($0\,\%$), partly covered ($< 50\,\%$), or covered ($\geq 50\,\%$). For each group, we report two quantities computed per image as in Eq. (2) and averaged over patients. The first is the change within the group using only its lesions. The second is the group's contribution to the evaluation metric: the change if only its lesions changed while all other lesions kept their volume and uptake. Since changes of different lesions in an image partly cancel, the contributions add up to slightly more than the evaluation metric.

Per-lesion terms. Because the PET losses constrain the total volume and uptake of an image, we test whether per-lesion terms can prevent changes in different lesions from canceling. In the intended test-time setting, per-lesion versions of the volume, uptake, and mean-Jacobian terms are added to the PET terms during IO on the network trained without the losses. The predicted lesion mask is split into connected components. Each of the eight largest components with at least 20 voxels gets its own relative volume and uptake change and mean Jacobian determinant, and the terms are averaged over the components weighted by size. Their weight is selected on the validation set with the selection rule of Sec. 3.3 applied to the median per-lesion $|\Delta\mathrm{MTV}| +$

$|\Delta\mathrm{TLG}|$, the quantity these terms target. The tested weights are listed in Appendix 13. As a separate diagnostic, we also apply the terms at their base weight to components of the provided lesion masks, which are unavailable in the intended test-time setting.

Contribution of each PET term. On the network trained without the losses, IO with all four PET terms is compared with IO in which one term at a time is switched off. This test-set comparison is a diagnostic analysis and is not used to select a new loss configuration.

Rigidity penalties. This comparison tests whether the penalties preserve bone geometry and whether they provide additional PET preservation beyond the PET-specific terms. On the network trained with the PET losses, IO with the PET terms is combined with no rigidity, Staring, Staring with erosion, or the per-bone rigid fit. Because the penalties have different units, a reference weight is first calibrated for each on five validation pairs so that its initial penalty value equals that of the rigid fit at $\lambda_{\mathrm{rig}} = 0.25$. Each method is then evaluated on the validation set over a common range of factors, giving a curve of CT Dice against within-bone distance distortion. For the test set, we choose the largest weight whose validation Dice is within two points of the variant without rigidity, using the same tolerance as above and capping the grid at $\times 100$. This gives $\times 3$ for Staring and $\times 100$ for Staring with erosion and the rigid fit. The calibrated reference weights and full sweep are listed in Appendix 13.

4 Results

Unless stated otherwise, results are reported on the test set (28 patients, 38 pairs); values are mean$\pm$SD over patients, and differences are given in percentage points. Validation data are used only for choices made before test evaluation and are identified explicitly. Except where stated, IO uses predicted masks, while the reported alignment and PET-preservation metrics use the provided labels. We first ask whether the losses alter the balance between alignment and image-level PET change, and whether they should act during training or only during IO. We then compare the selected configuration with baselines and investigate why residual image-level and per-lesion changes remain. Finally, we evaluate bone-rigidity penalties as an additional PET-preservation strategy and as a constraint on within-bone geometry.

4.1 Effect of the PET-Preservation and Rigidity Losses

We first test whether the PET-preservation and bone-rigidity losses reduce registration-induced changes in MTV and TLG and what they cost in alignment accuracy. Figure 2 compares four configurations (the full results with significance tests are in Table 3, Appendix A): no PET-preservation or bone-rigidity loss, PET preservation alone, rigidity alone, and both. Each network is evaluated before IO (squares; $\times$ without losses) and after IO (filled circles; $+$ without losses). In IO, the PET-preservation and bone-rigidity losses are enabled or disabled to match that network's training configuration. Without IO (squares vs. $\times$), training with the PET losses lowered the volume change of the lesions from 10.3 % to 8.2 % (significant) and the uptake change from 12.5 % to 9.8 % (not significant), at the cost of 1.9 Dice points. IO raised the Dice of every network by 3.7 to 4.7 points (each square to the filled circle of the same color, $\times$ to $+$). With the PET losses in training and IO (blue filled circle vs. $+$), $|\Delta\mathrm{MTV}|$ was 8.2 % instead of 11.3 % and $|\Delta\mathrm{TLG}|$ 10.3 % instead of 14.5 %, for 2.5 Dice points; with IO, these reductions were not significant. The rigidity loss alone

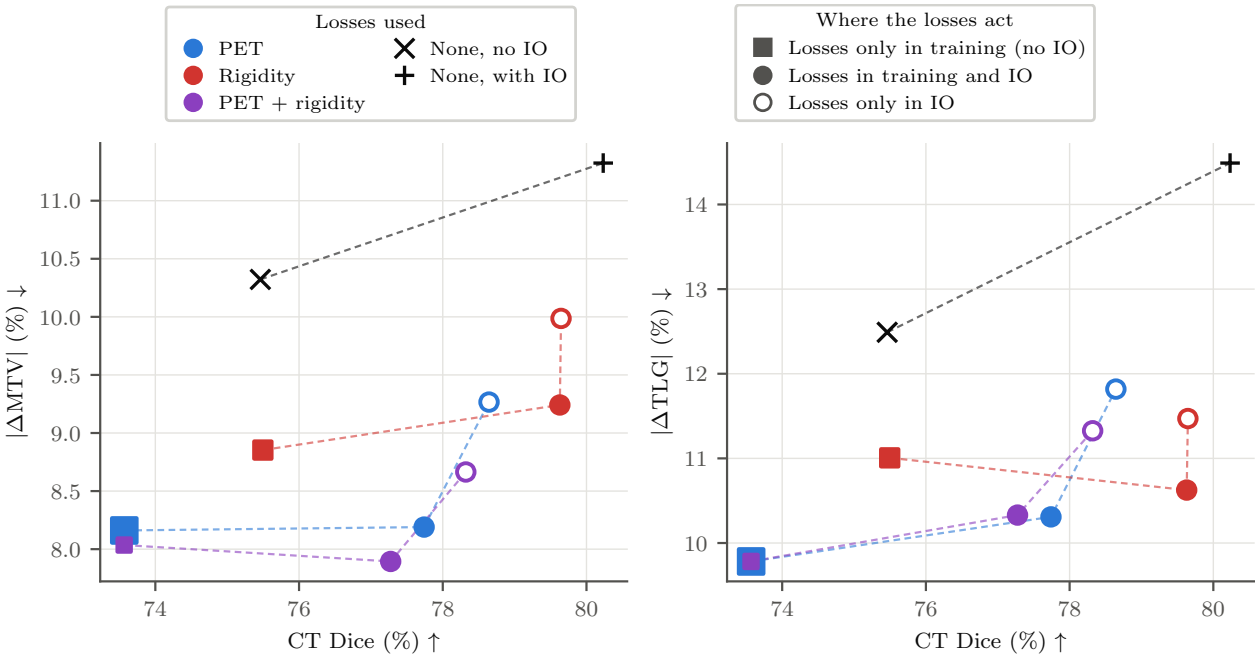


Fig 2 Effect of the PET-preservation and rigidity losses and of where they act on the test set (CT Dice: 28 patients, 38 pairs; metabolic tumor volume [MTV] and total lesion glycolysis [TLG]: 26 patients, 30 pairs). Color identifies the loss group; markers identify where the losses act (square: only in network training, shown before IO; filled circle: in training and IO; hollow circle: only in IO, on the network trained without losses). Without losses: network output ($\times$) and with IO ($+$). Lines connect configurations of the same loss group. Each point is the mean over patients after averaging each patient's eligible pairs; uncertainty and significance tests are reported in Table 3.

(red) did not change the Dice of the network and cost 0.6 points with IO; it lowered $|\Delta\mathrm{MTV}|$ significantly for the network output (10.3 % to 8.9 %) and $|\Delta\mathrm{TLG}|$ significantly with IO (14.5 % to 10.6 %). Combining both groups (purple) gave mean MTV and TLG changes close to those with the PET losses alone and was the least accurate variant (73.6 % and 77.3 % Dice). HD95 followed the Dice in every case (network 7.2 to 7.4 mm, with IO 6.6 to 7.1 mm). Among the matched configurations in this comparison, all network outputs and the IO variants without the PET losses stayed below the NDV threshold of 0.005 % (50 ppm) on every pair. With the PET losses in IO, the PET-only and combined configurations each exceeded it on one pair (150 and 262 ppm), while their patient-mean NDV values were at most 10 ppm.

The losses therefore change the accuracy–preservation balance; the next question is whether they need to shape the trained network or can act only during per-pair refinement.

4.2 Losses in Training or Only in IO

To test whether the losses need to act during training, Fig. 2 compares two variants for each loss group. Filled circles denote the network trained with the losses followed by IO with them, while hollow circles denote the network trained without them followed by the same IO. The second dashed segment of each color connects these variants. With both loss groups (purple), applying them only in IO increased Dice by 1.0 points (95 % CI 0.4 to 1.6) compared with using them in training and IO. The paired differences (IO only minus training and IO) were $+0.8$ points in $|\Delta\mathrm{MTV}|$ (95 % CI -0.4 to 2.0) and $+1.0$ points in $|\Delta\mathrm{TLG}|$ (-0.3 to 2.6). With PET losses

alone (blue), the Dice gain was 0.9 points (0.1 to 1.7), while $|\Delta\mathrm{MTV}|$ and $|\Delta\mathrm{TLG}|$ increased by 1.1 (-0.3 to 2.6) and 1.5 (-0.0 to 3.4) points. With rigidity alone (red), the paired differences were 0.0 (-0.0 to 0.1) Dice points, $+0.7$ (-0.0 to 1.8) MTV points, and $+0.8$ (0.0 to 1.9) TLG points. None of the preservation differences was significant after Holm correction; the intervals do not establish equivalent preservation. Without IO, training with the losses lowered the mean registration-induced PET changes in the network-only variants tested; detected reductions were limited to MTV with the PET losses, rigidity, or both (Fig. 2, squares; Table 3). The IO-only placement with both loss groups is therefore used as our selected method in the comparison with baselines.

4.3 Comparison with Baselines

To place our method in context, we compare it with CT-only registration baselines, with IO applied directly after affine pre-registration, and with our other variants (Table 1). Our method, with the losses only in IO, reached 78.3 % CT Dice and 6.89 mm HD95. Both were significantly better than NiftyReg (71.8 %, 7.88 mm), ConvexAdam (60.3 %, 9.32 mm), and the affine pre-registration (46.4 %, 14.00 mm). Its $|\Delta\mathrm{MTV}|$ (8.7 %) and $|\Delta\mathrm{TLG}|$ (11.3 %) did not differ significantly from those of NiftyReg (10.7 %, 12.6 %) or ConvexAdam (8.5 %, 9.3 %). Compared with the same network before IO, our method gained 2.9 Dice points (95 % CI 2.3 to 3.4). The paired changes in $|\Delta\mathrm{MTV}|$ and $|\Delta\mathrm{TLG}|$ were -1.7 points (95 % CI -3.6 to 0.3) and -1.2 points (-3.6 to 1.2), respectively, so an increase in image-level PET change was not detected. The NDV summaries in Table 1 are means over patients, whereas threshold exceedances count individual registration pairs. NiftyReg, tuned on the validation set, exceeded the folding threshold on 8 of 38 validation pairs and 13 of 38 test pairs (34 %), with up to 0.51 % NDV, and folded significantly more than our method; ConvexAdam stayed below the threshold on every pair. Our method exceeded it on two test pairs (76 and 125 ppm), and the variant with the losses also in training on one (262 ppm). IO directly after the affine pre-registration reached only 56.6 % Dice: within the same iteration budget, IO cannot replace the network. Even the affine pre-registration changed MTV by 4.5 % and TLG by 4.4 %, the lowest values among the registered images.

Among our variants, IO without the losses was the most accurate (80.2 % Dice, significantly higher than our method) but changed the lesions most ($|\Delta\mathrm{MTV}|$ 11.3 %, $|\Delta\mathrm{TLG}|$ 14.5 %). The losses only in IO lowered these changes to 8.7 % and 11.3 % at a cost of 1.9 Dice points. Using the losses in training and IO lowered them to 7.9 % and 10.3 % at a cost of 2.9 points. After correction for multiple comparisons, none of these preservation differences was significant. By the challenge's composite score, IO without the losses ranked first, our method second, and the variant with the losses in training and IO third, with overlapping bootstrap rank intervals (1–4, 1–4, and 1–5). The ranking was decided by accuracy: apart from the affine pre-registration on $|\Delta\mathrm{TLG}|$, no method won a significant comparison in preservation, and none in folding after the ranking's equivalence threshold. The network alone needs 11 s per pair, as fast as ConvexAdam and more than ten times faster than NiftyReg (128 s); IO adds about 2.5 minutes. Figure 3 illustrates anatomical alignment in a test pair selected for its tenfold between-image change in lesion volume (from 2400 to 24 191 voxels). The images were initially far apart (CT Dice 3.5 %); our method aligned them to 83.7 % Dice (NiftyReg 72.3 %, ConvexAdam 64.5 %), and NDV remained below the challenge threshold. Before registration, the CT overlay (Fig. 3c) and label residual (e) show misalignment across the whole trunk. After registration (d, f), the residual is reduced to thin

Table 1 Comparison with baselines on the test set (28 patients, 38 follow-up→baseline pairs; PET: 26 patients, 30 pairs). Values are mean±SD over patients (a patient's eligible pairs are averaged first). HD95: 95th-percentile Hausdorff distance; MTV: metabolic tumor volume; TLG: total lesion glycolysis; NDV: non-diffeomorphic volume. Rank: descriptive challenge significance ranking, with the 95 % interval of the rank over 1000 patient bootstrap samples in brackets, over all methods except Initial, which is a reference (its MTV/TLG change is 0 by construction because nothing is resampled). Training / IO: PET and rigidity losses (PET + rig.) used in training and in instance optimization (IO); IO "–": no IO, "basic": IO with the similarity and regularization terms only. †/‡: Ours + IO with PET + rig. only in IO (our method, chosen on the validation set) is significantly better/worse than this method (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected over nine comparisons for CT/HD95/NDV and eight for PET within each column, $p < 0.05$; NDV tested after the 0.005 % equivalence threshold). NDV is summarized as mean±SD over patients; $>$ thr. is the percentage of individual pairs with %NDV above 0.005 %. Runtime per pair on one NVIDIA RTX A6000, including the affine step (our methods: measured in earlier runs with the same settings; NiftyReg: measured on the validation pairs). Affine + IO: our instance optimization directly after the affine pre-registration, without the network. Bold: best mean among ranked methods.

Method	Training	IO	Rank ↓	Dice (%) ↑	HD95 (mm) ↓	$\lvert\Delta\mathrm{MTV}\rvert$ (%) ↓	$\lvert\Delta\mathrm{TLG}\rvert$ (%) ↓	NDV (ppm) ↓	$>$ thr. (%)	Runtime (s)
Initial	–	–	–	9.5±8.1†	44.37±21.32†	0.0±0.0	0.0±0.0	0±0	0	–
Affine	–	–	8 [3–9]	46.4±11.0†	14.00±4.83†	**4.5**±4.6	**4.4**±4.6‡	**0**±0	0	10.4±0.3
NiftyReg	–	–	6 [2–8]	71.8±4.2†	7.88±2.66†	10.7±16.3	12.6±21.4	349±1000†	34	128.3±23.4
ConvexAdam	–	–	7 [6–9]	60.3±5.6†	9.32±2.59†	8.5±7.5	9.3±9.8	**0**±0	0	11.4±1.2
Affine + IO	–	PET + rig.	8 [4–9]	56.6±12.1†	12.88±4.81†	5.3±4.1	6.8±4.0	0.45±0.31	0	155.5±2.6
Ours	–	–	4 [3–6]	75.5±4.7†	7.20±2.30†	10.3±9.5	12.5±11.1	0.056±0.057	0	11.0±0.4
Ours	PET + rig.	–	5 [1–7]	73.6±4.6†	7.44±2.37†	8.0±9.0	9.8±10.9	0.64±0.62	0	11.3±0.3
Ours + IO	–	basic	1 [1–3]	**80.2**±4.2‡	**6.64**±2.28‡	11.3±11.9	14.5±14.1	0.65±0.13	0	153.9±2.6
Ours + IO	PET + rig.	PET + rig.	3 [1–5]	77.3±4.9†	7.07±2.38†	7.9±9.7	10.3±10.4	10.4±49.3	3	156.6±2.7
Ours + IO	–	PET + rig.	2 [1–4]	78.3±4.4	6.89±2.31	8.7±10.7	11.3±11.2	4.15±9.45	5	157.2±2.6

bands at organ boundaries and parts of the lumbar vertebrae. The warped lesion contours lie in the corresponding bones (h). The Jacobian determinant (g) shows fine-grained changes in the local volume scale of the sampling transform throughout the body, which IO adds to the smooth network field because it has no smoothness term ($\lambda_{\mathrm{smooth}} = 0$).

The remaining analyses examine why measured PET changes persist after loss-guided IO.

4.4 Accuracy–Preservation Trade-off

In a post-hoc test-set sensitivity analysis, we vary the PET-loss weights in IO to show how alignment and lesion preservation change together and whether stronger weights resolve the residual error. Figure 4 varies the PET weights in IO on the network trained without the losses without reselecting the method. The smallest factor ($\times 10^{-4}$) lowered the volume change of the predicted lesion masks from 9.3 % (IO without the PET losses) to 2.1 % and that of the provided lesions from 11.3 % to 9.3 %. Mean Dice remained 80.2 %, while $\lvert\Delta\mathrm{TLG}\rvert$ fell from 14.5 % to 12.9 %. Larger factors reduced the change of the predicted masks further, to below 1 % from $\times 0.01$ on, but not that of the provided lesions (9.0 to 9.8 %), while Dice decreased steadily, to 78.6 % at $\times 0.1$ and 78.0 % at $\times 1$. Compared with the network output before IO (dashed), IO with the PET losses gained 3.2 Dice points and had slightly lower mean changes ($\lvert\Delta\mathrm{MTV}\rvert$ 9.3 % vs. 10.3 %, $\lvert\Delta\mathrm{TLG}\rvert$ 11.8 % vs. 12.5 % at $\times 0.1$). Neither PET difference was significant (two-sided Wilcoxon signed-rank test, uncorrected $p = 0.17$ and $p = 0.86$; lower for 14 and 13 of 26 patients). The original validation grid contained no nonzero factor below $\times 0.1$. In this test-set sweep, the selected factor $\times 0.1$ therefore cost 1.6 Dice points more than $\times 10^{-4}$ for the same volume change (9.3 %) and 1.1

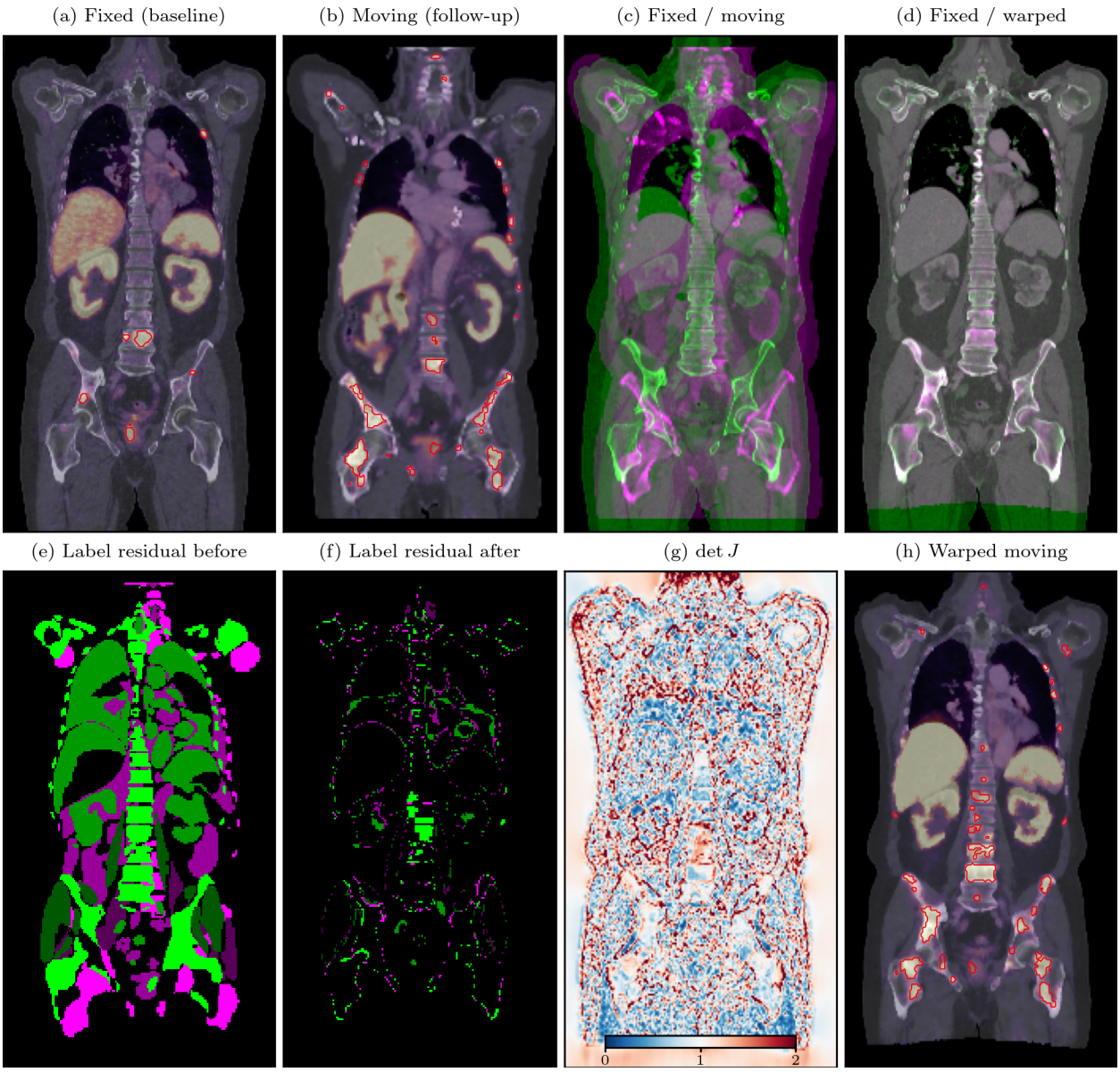


Fig 3 Anatomical alignment example from the test set, registered by our method (PET-preservation and rigidity losses only in instance optimization [IO]); one coronal slice. Patient 0146 was selected for its large between-image change in lesion volume (2400 voxels at baseline and 24 191 at follow-up). The example illustrates anatomical alignment and deformation. (a) Fixed and (b) moving CT with PET (SUV 0–8, 35 % opacity) and provided lesion contours (red). (c, d) CT of the fixed image (green) overlaid with the moving image (magenta) before (c) and after (d) registration; gray where they agree. (e, f) Organ labels that differ before (e) and after (f) registration: green where the fixed image has a label that the moving image lacks there, magenta where only the moving image is labeled; bright, medium, and dark shades for bones, organs, and muscles. (g) Jacobian determinant of the fixed-to-moving sampling transform: white is volume-preserving (1), blue is below 1 and red is above 1; changes in the displayed warped anatomy have the opposite direction. (h) Warped moving CT with PET and warped lesion contours.

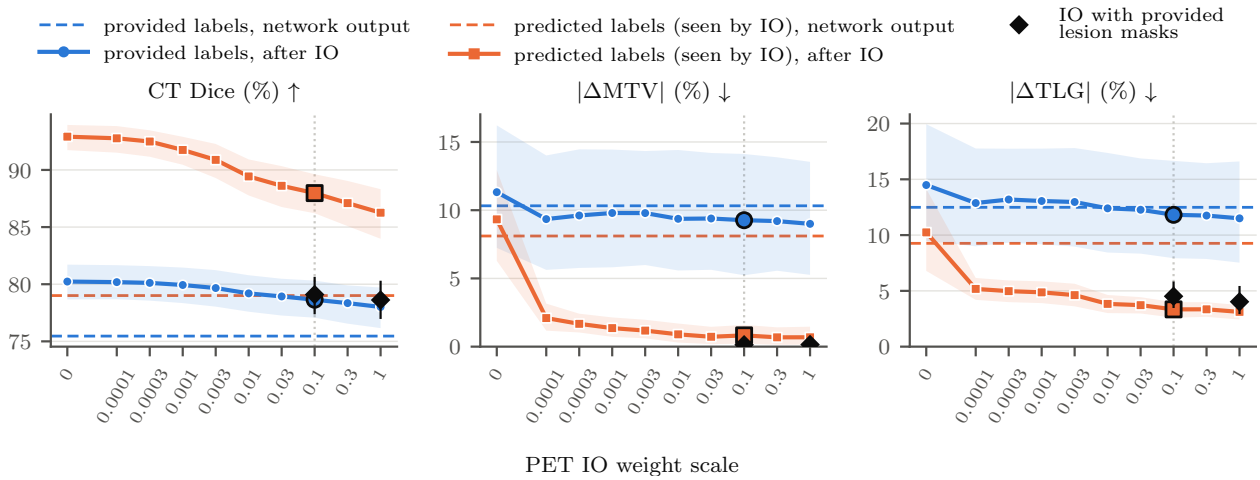


Fig 4 Effect of the PET-loss weight in instance optimization (IO) on the test set: the network trained without PET losses (T1) is followed by IO with all PET weights scaled by the given factor (0: PET losses off; 1: training-derived weights). Metrics are computed on the predicted labels optimized by IO (orange squares; organs: TotalSegmentator, lesions: nnU-Net) and on the provided evaluation labels (blue circles). CT Dice includes 28 patients and 38 pairs; metabolic tumor volume (MTV) and total lesion glycolysis (TLG) change include the 26 patients and 30 pairs with lesions in both label sets. Solid: after IO; dashed: network output before IO, the same for every factor. Outlined point and dotted line: operating point tuned on the validation set. Black diamonds: IO with the provided instead of the predicted lesion masks (at $\times 0.1$ and $\times 1$), evaluated on the provided labels. Curves show means over patients after averaging eligible pairs; bands show 95 % patient-bootstrap confidence intervals.

points less uptake change (11.8 %). Because this comparison was made on the test set, it does not justify replacing the validation-selected factor. On the predicted masks, $|\Delta\mathrm{TLG}|$ fell from 10.2 % to 5.2 % at $\times 10^{-4}$ and to about 3 % from $\times 0.1$ on. The same gap between the labels IO optimizes and the provided labels appeared for the organs: without the PET losses, IO reached 92.9 % Dice on the predicted organ labels but 80.2 % on the provided labels, starting from 79.0 % and 75.5 % for the network output.

In the diagnostic run with provided lesion masks (black diamonds), the measured changes fell from 9.3 % to 0.1 % for MTV and from 11.8 % to 4.5 % for TLG at the tuned factor; Dice rose from 78.6 % to 79.1 %. At $\times 1$, the changes were similar (0.2 % MTV, 4.0 % TLG; 78.6 % Dice), so a larger weight hardly reduced the remaining uptake change even with provided masks. The near-zero volume change is that of the image total, which this run optimizes directly; the change of individual lesions remained (Sec. 4.6). This gap between optimization on predicted masks and evaluation on provided labels motivates an analysis of lesion coverage.

4.5 Lesion Coverage in Predicted Masks

To locate the remaining registration-induced change, we group all 1435 provided lesions by their overlap with the mask used by IO (Fig. 5). This analysis includes the 108 lesions smaller than five voxels. Of all lesions, 214 had no overlap (missed; 6 % of lesion volume), 155 had less than 50 % overlap (partly covered; 6 %), and 1066 had at least 50 % overlap (covered; 88 %).

PET IO reduced the change of covered lesions below that of the network output, from 9.1 % to 6.0 % for $|\Delta\mathrm{MTV}|$ and from 11.0 % to 8.4 % for $|\Delta\mathrm{TLG}|$ (IO without and with the PET losses at the tuned weights; network output 7.9 % and 9.2 %). Partly covered lesions changed as much

with the PET losses as without them (12.5 % and 12.0 %), and missed lesions less (14.4 % and 17.0 %) but still more than at the network output (12.4 %). The loss has no direct target for a missed lesion and only a partial target for a partly covered lesion. After PET IO, missed lesions therefore contributed more than half of the remaining change: 5.2 of 9.3 points of $|\Delta\mathrm{MTV}|$ and 6.2 of 11.8 points of $|\Delta\mathrm{TLG}|$, although they make up only 6 % of the lesion volume. Covered lesions contributed 3.3 and 5.4 points. Scaling the PET weights from $\times 0.1$ to $\times 1$ changed none of these values by more than 0.3 points.

Without the PET losses, IO increased the change of every group over the network output, for covered lesions from 7.9 % to 9.1 % ($|\Delta\mathrm{MTV}|$) and from 9.2 % to 11.0 % ($|\Delta\mathrm{TLG}|$), and the overall change from 10.3 % to 11.3 % and from 12.5 % to 14.5 %. This agrees with the per-lesion analysis (Sec. 4.6), in which lesions had higher volume changes after IO without the losses. In the diagnostic run with provided lesion masks, the contribution of lesions that nnU-Net missed fell from 5.2 to 1.2 points of $|\Delta\mathrm{MTV}|$ and from 6.2 to 2.1 points of $|\Delta\mathrm{TLG}|$, and that of covered lesions from 3.3 to 0.9 and from 5.4 to 3.8 points. The near-zero overall change with provided masks does not mean that individual lesions kept their volumes. If only the missed, partly covered, or covered lesions had changed, MTV would have changed by 1.2, 1.2, and 0.9 points, respectively. Some lesions therefore grew while others shrank, and their changes canceled in the image total measured by both the PET loss and the evaluation metric (see also Sec. 4.6).

As a post-hoc attempt to increase lesion coverage, we combined the nnU-Net predictions with voxels above SUV 3 outside a 10-mm halo around predicted organs with physiological uptake, following the recall-biased approach of Dey and Dalca.[30] On the moving test images with lesions (mean over patients), lesion detection increased from 74 % to 91 %, while voxel-level precision fell from 0.71 to 0.22. Compared with IO using the nnU-Net mask, MTV change was 10.3 % instead of 9.3 %, TLG change was 12.5 % instead of 11.8 %, and CT Dice was 79.3 % instead of 78.6 %. None of these differences was significant (two-sided Wilcoxon signed-rank test, uncorrected $p \geq 0.09$). The per-lesion median was similar (13.5 % vs. 13.7 %). Thus, the tested recall-biased mask produced no detected reduction in either image-level or per-lesion change despite detecting more lesions.

4.6 Per-Lesion Volume Change

Mask coverage explains part of the residual image-level change, but image totals can also hide opposing changes in different lesions. To reveal these changes, Fig. 6 shows the volume change of 1327 lesions from 26 patients, after excluding 108 components smaller than five voxels. Small lesions changed most in every condition: for bone-associated lesions with IO and without the losses (+), the median change was 30 % (small), 19 % (medium), and 10 % (large). A small lesion has fewer than about 20 voxels, so gaining or losing one or two voxels is already a large relative change. The per-lesion uptake change followed the volume change in all strata (Fig. 10, Appendix B). Without the losses, IO increased lesion distortion ($\times$ to +), especially for small lesions (soft tissue from 17 % to 33 %, bone-associated from 25 % to 30 %). Bone-associated lesions tended to grow (median signed change +1 to +5 %), whereas soft-tissue lesions tended to shrink (-3 to -14 %). The rigidity loss (red) acted mainly on bone-associated lesions, as expected from its construction. With IO, it lowered their median change for small, medium, and large lesions from 30 % to 20 %, 19 % to 12 %, and 10 % to 5 %. It left medium and large soft-tissue lesions unchanged at 19 % and 17 %. The PET losses (blue) acted in both locations, already without IO (squares), and with IO, for

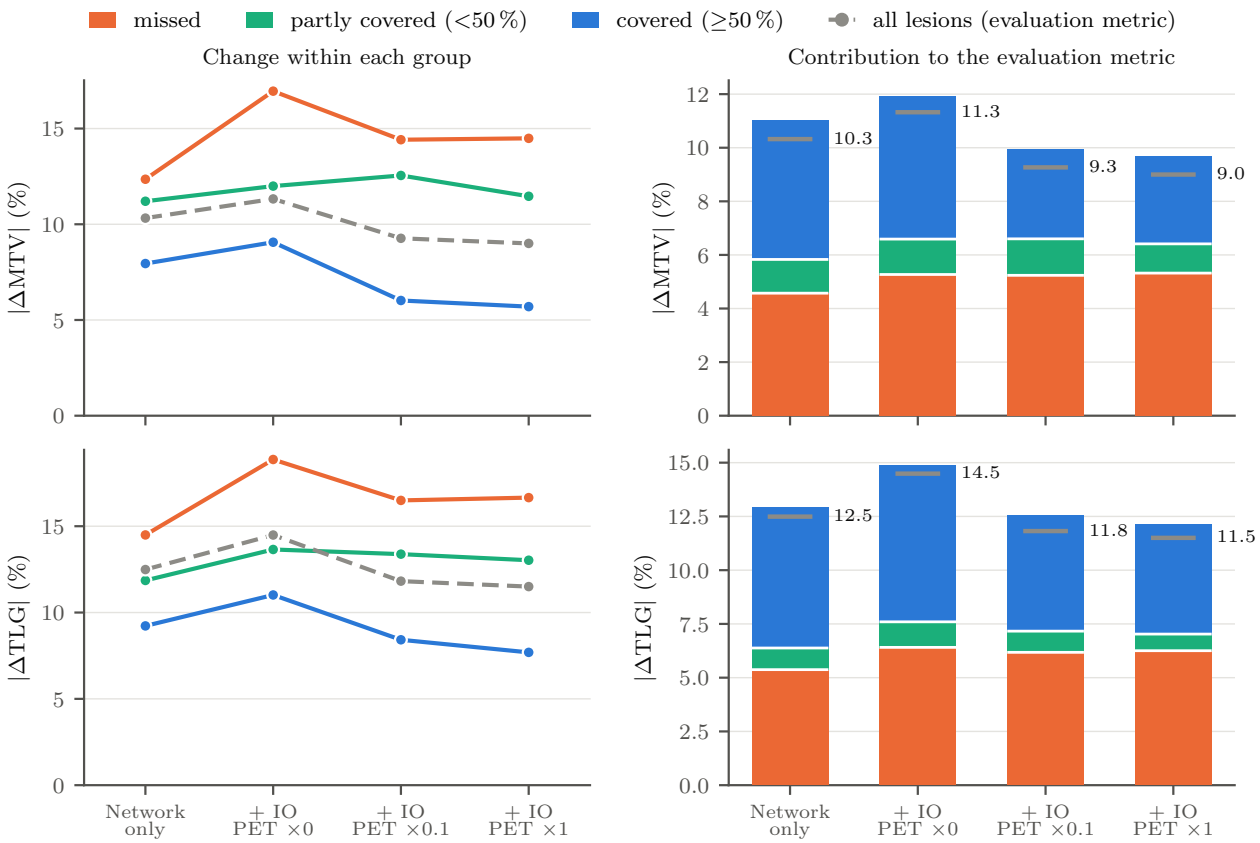


Fig 5 Registration-induced metabolic tumor volume (MTV) and total lesion glycolysis (TLG) change of all 1435 provided lesion components in 26 test patients (30 pairs), including components smaller than five voxels. Components are grouped by coverage from the predicted lesion mask used by instance optimization (IO): missed (214 lesions, 6 % of volume), partly covered (155 lesions, 6 %), and covered (1066 lesions, 88 %). The network is trained without PET losses; IO uses the PET weights scaled by the stated factor. Colors identify coverage groups. Left: absolute change within each group, computed per image from the group's summed volume or uptake and averaged over patients with lesions in that group; gray dashed: all lesions, the evaluation metric. Right: contribution of each group to the image-level evaluation metric, averaged over patients; gray ticks show the evaluation metric and lie below the stacked total when opposing lesion changes cancel.

example from 19 % to 10 % for medium soft-tissue lesions and from 30 % to 16 % for small bone-associated lesions. With both groups (purple), bone-associated lesions changed by 14 %, 9 %, and 6 %, similar to the PET losses alone; only for large bone-associated lesions, the rigidity loss alone changed them less (5 %). With the losses, IO did not reduce the per-lesion change below that of the network output (circles vs. squares of the same color), while no image-level preservation difference was detected either (Sec. 4.1). The PET losses in IO penalize the change in the total volume and uptake of the predicted lesion mask, so the growth of some lesions can offset the shrinkage of others. Even with the provided lesion masks in IO, which lowered the image-level $|\Delta\mathrm{MTV}|$ to 0.1 % (Sec. 4.4), the median per-lesion change was 13.5 %, as with the predicted masks (13.7 %).

We next tested whether terms on individual predicted components could prevent this cancellation in the intended test-time setting. On the validation set, the selection chose ten times the base weight and reduced the median per-lesion $|\Delta\mathrm{MTV}|$ from 12.5 % to 11.3 %. Adding the validation-selected per-lesion terms did not provide a consistent benefit on the test set. The median per-lesion

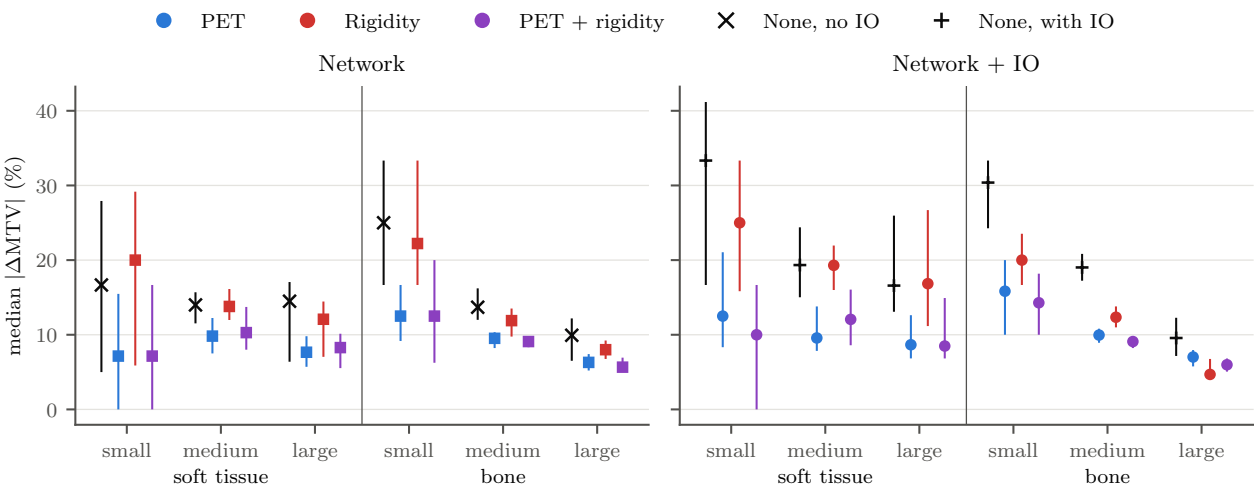


Fig 6 Per-lesion absolute volume change on the test set (1327 lesions from 26 patients and 30 pairs after excluding 108 components smaller than five voxels), stratified by location (soft tissue or bone-associated, with $\geq$50 % of the lesion on bone) and size (small $<$ 0.5 ml, medium 0.5 to $<$ 5 ml, large $\geq$ 5 ml). Points show the median $|\Delta\mathrm{MTV}|$ per lesion; left: network output, right: instance optimization (IO) with the same losses. Within each size stratum, the loss groups are ordered PET, rigidity, and PET plus rigidity; squares and circles denote without and with IO, while $\times$ and $+$ denote the corresponding configurations without losses. Error bars: 95 % confidence intervals from a patient-cluster bootstrap.

volume change fell only from 13.7 % to 13.3 % and decreased for only 5 of 26 patients, while the median for small lesions was higher (20.0 % vs. 16.7 %). The mean image-level $|\Delta\mathrm{MTV}|$ remained at 9.3 %, $|\Delta\mathrm{TLG}|$ changed from 11.8 % to 12.0 %, and Dice rose by 0.6 points. In the separate diagnostic using provided lesion masks, the terms at their base weight also left the median per-lesion change unchanged at 13.5 %. The medians of small lesions take only a few distinct values (e.g. 12.5 %, 14.3 %, 33.3 %) because they are ratios of small voxel counts.

4.7 Contribution of Each PET Term

To identify which constraints account for preservation in IO, we remove one PET term at a time from IO on the network trained without the losses (Table 2). The mean-Jacobian term $\mathcal{L}_{\overline{J}}$ carried the preservation: without it, $|\Delta\mathrm{MTV}|$ increased from 9.3 % to 10.4 % and $|\Delta\mathrm{TLG}|$ from 11.8 % to 13.5 % (both significant). Without the voxel-wise term $\mathcal{L}_J$, $|\Delta\mathrm{TLG}|$ increased to 12.3 % (significant), whereas $|\Delta\mathrm{MTV}|$ was unchanged (9.3 %). The two direct measurement terms did not improve the evaluated endpoints in this test-set ablation. Without $\mathcal{L}_{\mathrm{MTV}}$, Dice increased from 78.6 % to 79.3 % and $|\Delta\mathrm{TLG}|$ decreased to 10.7 % (both significant; $|\Delta\mathrm{MTV}|$ 8.6 %, not significant). Without $\mathcal{L}_{\mathrm{TLG}}$, $|\Delta\mathrm{TLG}|$ decreased to 9.8 % (significant), while Dice (78.7 %) and $|\Delta\mathrm{MTV}|$ (8.6 %) did not change significantly. Thus, the test-set results do not support a claim that every PET term improves the evaluated preservation metrics. One possible explanation is that both direct terms measure change on the linearly warped, soft mask and PET (Eq. 4), whereas the evaluation warps both by nearest neighbor. The ablation is consistent with this surrogate mismatch but does not establish it as the cause. Because the four-term configuration was fixed before test evaluation, this diagnostic ablation was not used to select a reduced loss.

Table 2 Diagnostic contribution analysis of each PET term in instance optimization (IO) on the test set (CT Dice: 28 patients, 38 pairs; metabolic tumor volume [MTV] and total lesion glycolysis [TLG]: 26 patients, 30 pairs): network trained without PET losses (T1), followed by IO with all PET terms or with one term removed. Mean±SD over patients. †/‡: significantly better/worse than with all terms (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected over four comparisons within each column, $p < 0.05$). This test-set analysis was not used to select a reduced loss.

IO PET terms	Dice (%) ↑	$\lvert\Delta\mathrm{MTV}\rvert$ (%) ↓	$\lvert\Delta\mathrm{TLG}\rvert$ (%) ↓
All PET terms	78.6±4.4	9.3±11.6	11.8±11.7
without $\mathcal{L}_{\mathrm{MTV}}$	79.3±4.6†	8.6±11.6	10.7±11.9†
without $\mathcal{L}_{\overline{J}}$	78.7±4.5	10.4±11.1‡	13.5±11.2‡
without $\mathcal{L}_{\mathrm{TLG}}$	78.7±4.4	8.6±11.7	9.8±12.5†
without $\mathcal{L}_{J}$	78.6±4.5	9.3±11.4	12.3±11.5‡

4.8 Rigidity Penalties

Bones should retain their internal geometry during registration even when they move relative to one another. Because label overlap does not measure this property, we compare rigidity penalties by how well they preserve within-bone distances and report the accompanying changes in bone and soft-tissue alignment. In the finite-difference Staring penalty, 21.2 % of the stencil reads leave the bone of the center voxel across all 560 labeled challenge images. Eroding the bones removes this leakage but retains only 35.5 % of the bone voxels and removes 23 % of the bones entirely. For ribs, it retains 2.4 % of the voxels and removes 58 % of the structures.

On the validation set (Fig. 7), the per-bone rigid fit reached a lower within-bone distortion than the Staring penalty at every Dice level, most clearly for long distances, where Staring did not get below 2.8 % while the rigid fit reached 1.8 %. With erosion, the Staring penalty barely changed the distortion even at $\times 100$ (short distances 8.7 % vs. 12.7 % without rigidity), and Dice stayed almost constant, because most of the skeleton is no longer constrained. Within the two-point tolerance, the selection chose $\times 3$ for Staring and the largest grid value, $\times 100$, for the other two penalties.

On the test set, Fig. 8a shows each rigidity measure as a percentage of its value with IO without rigidity (dashed line at 100 %). Each penalty produced the lowest value for the measure it optimizes: the rigid fit lowered the rigid-fit residual to 53 %, and Staring lowered the Staring energy to 18 %. Neither penalty directly optimizes distance distortion. The rigid fit nevertheless produced the lowest values: 41 % (short) and 42 % (long) of the distortion without rigidity, compared with 47 % and 73 % for Staring and 69 % and 80 % for Staring with erosion. Fig. 8b shows the cost in alignment. The rigid fit and Staring lowered CT Dice by 1.6 and 2.0 points and bone Dice by about 4 points. Soft-tissue Dice was unchanged with Staring and rose by 0.9 points with the rigid fit. Staring with erosion cost almost no Dice but also constrained the bones least.

In absolute terms (Table 8, Appendix F), the rigid fit reduced the short-distance distortion from 13.4 % to 5.0 % and the long-distance distortion from 4.4 % to 1.8 %. All three penalties improved every rigidity measure significantly over IO without rigidity. All pairwise distortion differences between the penalties were also significant. Staring also removed folding inside bones (NDV in bone from 6.5 to 0.3 ppm), whereas the rigid fit, which constrains only the displacement, left it unchanged (9.6 ppm, not significant).

When added on top of the PET losses, none of the three rigidity penalties produced a detected additional change in lesion volume or uptake preservation on the test set (Table 8). This comparison

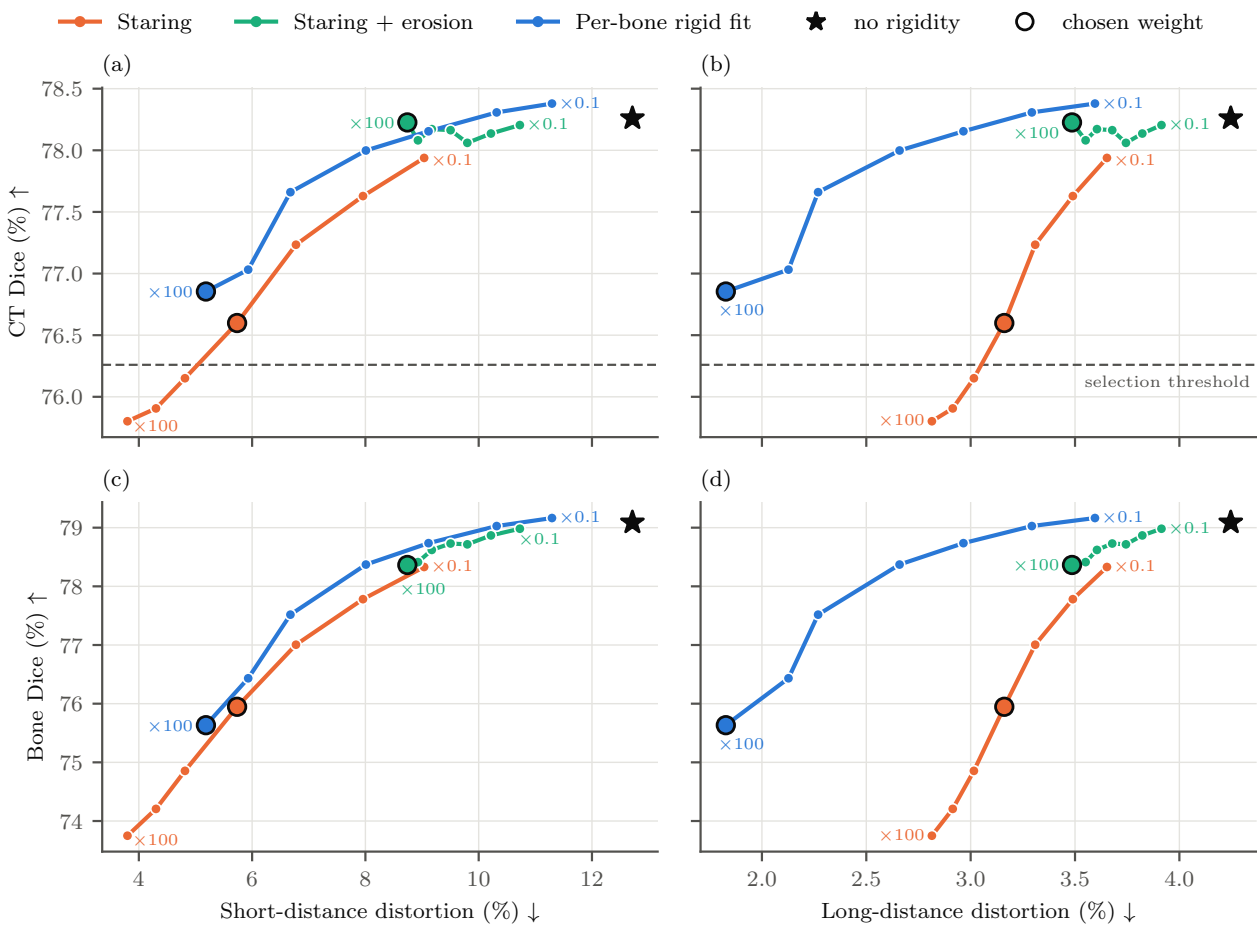


Fig 7 Rigidity-weight sweep on the validation set (26 patients, 38 pairs): network trained with the PET losses, followed by instance optimization (IO) with the PET terms and one rigidity penalty at 0.1 to 100 times its calibrated reference weight (factors at the ends of each line). Top: CT Dice; bottom: bone Dice; left: distortion of short (5–15 mm) and right: long ($\geq$ 15 mm) within-bone distances, defined as the mean absolute relative change in distance between point pairs in the same bone. Color identifies the rigidity penalty. Star: IO without rigidity; dashed line in (a, b): selection threshold two Dice points below the reference; outlined marker: weight chosen for the test set, the largest within the threshold up to $\times 100$. Points are means over patients after averaging each patient's pairs.

differs from Fig. 2, where rigidity used without the PET losses significantly reduced network-output $|\Delta\mathrm{MTV}|$ and IO $|\Delta\mathrm{TLG}|$ relative to the corresponding unconstrained variants. On the validation set (Fig. 9), every penalty and weight changed the lesions less than IO without rigidity, but the change did not decrease with increasing weight. The drop between weight 0 and the smallest weight came mostly from two patients whose volume change fell by 27 and 10 points with the rigid fit at $\times 0.1$. Without these patients, all weights remained within about one point of IO without rigidity ($|\Delta\mathrm{MTV}|$ 3.7 to 4.9 % vs. 4.4 %). Per lesion, the penalties lowered the median volume change of bone-associated lesions only from 9.3 % to 7.7–8.8 %, whereas an exactly affine transformation (the affine pre-registration) changed them by a median of 1.0 %.

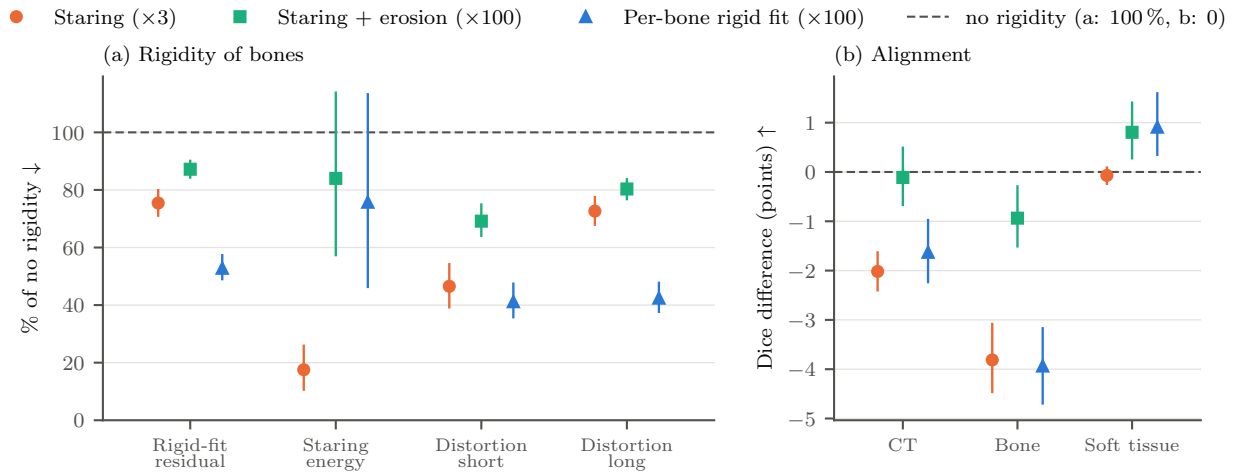


Fig 8 Rigidity penalties evaluated with all rigidity measures on the test set (28 patients, 38 pairs). The network is trained with the PET losses and followed by instance optimization (IO) with the PET terms and one rigidity penalty at the weight chosen on the validation set. (a) Rigidity of bones relative to IO without rigidity (dashed, 100 %), per patient: residual of a rigid fit to each bone (the quantity the rigid-fit penalty minimizes), Staring energy, i.e. the Staring penalty (properness, orthonormality, and affinity) on the whole bone (the quantity the Staring penalty minimizes), and the distortion of short (5–15 mm) and long (≥15 mm) within-bone distances (minimized by neither). (b) Change of CT, bone, and soft-tissue Dice relative to IO without rigidity. Color and marker shape identify the rigidity penalty. Points are means over patients after averaging each patient's pairs; bars are 95 % patient-bootstrap confidence intervals.

5 Discussion

The main result is a bounded practical choice: loss-guided IO improves alignment over network output when predicted masks and per-pair refinement time are available, but the present data do not establish better or equivalent image-level PET preservation. Applying the losses only during IO gives higher alignment accuracy than also using them in training, with an unresolved preservation trade-off. The remaining PET change reflects three limits of the method and evaluation: incomplete lesion masks, cancellation between lesions in image totals, and a mismatch between the optimized and evaluated quantities. Bone rigidity serves both as a PET-preservation strategy and as a constraint on within-bone geometry: rigidity alone reduced particular PET endpoints, but it provided no detected additional preservation benefit on top of the PET losses.

5.1 Choosing Where to Apply the Losses

Network-only registration takes about 11 s per pair on our GPU and requires only the image pair at inference. Loss-guided IO additionally requires predicted organ and lesion masks and about 2.5 minutes per pair. Its advantage in this comparison is additional alignment rather than demonstrated improvement in MTV or TLG preservation. The network output remains the practical choice when masks or per-pair runtime are unavailable.

Among the IO variants, applying the losses only during refinement supports an accuracy gain, but the paired intervals still allow worse MTV or TLG preservation than when the losses are also used in training. Without IO, training with the losses lowered the mean registration-induced changes in lesion volume and uptake; only the MTV reduction was detected after correction. Training with the losses may therefore remain relevant when many follow-up sessions must be registered without per-pair refinement.

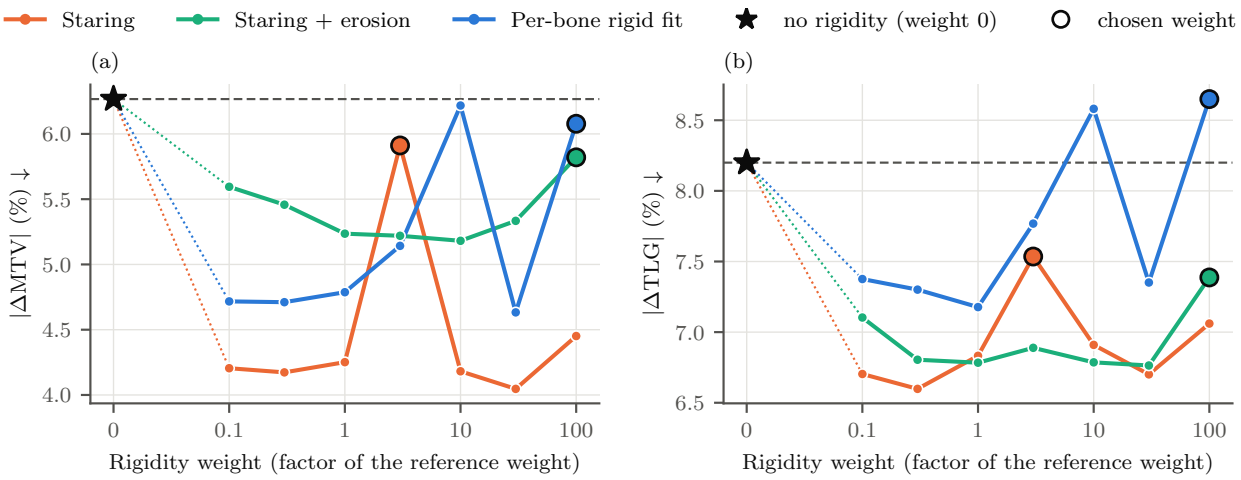


Fig 9 Effect of rigidity weight on lesion preservation on the validation set (26 patients, 32 lesion-positive pairs): network trained with the PET losses, followed by instance optimization (IO) with the PET terms and one rigidity penalty at 0.1 to 100 times its calibrated reference weight. (a) Absolute metabolic tumor volume (MTV) change; (b) absolute total lesion glycolysis (TLG) change. Color identifies the rigidity penalty. Star and dashed line: IO without rigidity (weight 0); dotted segment: connection to the smallest nonzero weight; outlined marker: weight chosen for the test set. Points are means over patients after averaging each patient's lesion-positive pairs.

One possible explanation for the IO-only result is how each stage locates lesions. During network training, lesion labels define the preservation losses, but the trained registration network receives only images at inference and must respond to lesion-related image cues without a mask. IO instead uses an explicit mask from a separately trained lesion-segmentation model, which can direct preservation to the predicted lesions in each pair. This may explain why adding the losses during training produced no detected preservation benefit once IO was applied. The comparison does not test what lesion information the registration network learns, and the segmentation model cannot by itself explain the Dice advantage of IO-only placement: both IO variants use the same predicted masks.

5.2 Limits of Lesion Preservation

Three mechanisms must be distinguished. First, incomplete lesion-mask coverage limits how far IO can reduce image-level MTV and TLG change. Second, image-level aggregation allows growth and shrinkage of different lesions to cancel. Third, the soft quantities optimized by the losses differ from the nearest-neighbor quantities used for evaluation. The contrast between near-zero volume change on the predicted masks and substantial change on the provided masks shows that IO can satisfy its objective without controlling the evaluation endpoint. Missed lesions were the largest measured contributor to the residual, although they represented only 6 % of lesion volume; covered lesions also contributed, so incomplete coverage is not the sole explanation. Their weight comes from patients with little tumor volume because the metric is relative per image and averaged over patients. In the third of patients with the smallest lesion volume, missed lesions made up 33 % of the volume, compared with 23 % and 3 % in the other thirds. This group caused 72 % of the missed-lesion contribution. The post-hoc recall-biased mask increased lesion detection but sharply reduced precision and did not improve measured preservation (Sec. 4.5). The additional false-positive regions may dilute the preservation constraints over voxels outside the lesions being

evaluated, so this experiment does not show that recall gains are ineffective when precision can be retained. Conversely, the diagnostic run with provided lesion masks nearly eliminated image-level MTV change while improving alignment. Together, these experiments identify mask quality as a major limit of image-level preservation, but the remaining TLG change shows that segmentation does not explain all residual error.

Because IO meets its objective on the predicted masks already at small weights, additional PET weight mainly costs alignment. The post-hoc sweep shows that the original validation grid did not sample the low-weight plateau, but a test-set observation cannot define a replacement operating point. A denser low-weight sweep on validation data followed by independent confirmation would be required before changing the selected weight.

Mask coverage does not explain the second limit: aggregate MTV and TLG changes hide how individual lesions are deformed. Small lesions are especially sensitive because gaining or losing one or two voxels is already a large relative change. The location-specific effects were consistent with the losses: rigidity acted mainly on bone-associated lesions, whereas the PET losses acted in both locations. Nevertheless, IO with the losses did not reduce per-lesion change below the network output, even with provided lesion masks. The PET losses constrain the total volume and uptake of an image, which can remain nearly unchanged while single lesions grow or shrink. Indeed, provided masks reduced image-level MTV change to 0.1 %, while the median per-lesion change remained 13.5 %.

The tested per-lesion terms did not solve this problem on the test set. They leave small and additional components unconstrained, depend on predicted components that may miss, merge, or split lesions, and use a soft interpolation surrogate that differs from the evaluation. Even the diagnostic with provided masks did not lower the median per-lesion change. An effective per-lesion objective may therefore need to include small lesions and more closely match the nearest-neighbor evaluation.

The leave-one-out experiment exposes the third mechanism: the optimized surrogate does not always track the evaluated endpoint. The Jacobian terms supported the measured preservation, whereas removing either direct measurement term improved evaluated TLG change. Both terms act on the soft, linearly warped mask, whose change differs from the nearest-neighbor change that is evaluated. This is one possible explanation for the result, but the experiment does not show that interpolation mismatch caused it. We retained all four terms because this leave-one-out analysis used the test set and the configuration had been fixed beforehand. A closer surrogate or a reduced set of terms would require selection on validation data and confirmation on independent data.

5.3 Interpreting the Evaluation

The significance-based ranking favored accuracy because alignment differences were more consistent across patients than differences in MTV and TLG. Our selection rule instead considered preservation among candidates within a fixed Dice tolerance (Sec. 3.3). This makes the trade-off explicit, but it does not establish which balance is preferable for a clinical task. The MTV and TLG endpoints compare the same follow-up acquisition before and after warping, so they isolate change introduced by registration rather than biological change between sessions. This study does not define or validate a threshold at which such change alters lesion tracking, response classification, or another clinical decision. The reported differences should therefore be interpreted as technical fidelity measures rather than demonstrated clinical benefit. Because image-level changes

allow cancellation between lesions, per-lesion changes should accompany them when the intended application follows individual lesions.

Both measured alignment and preservation also depend on the labels: IO fits predicted masks more closely than the independent evaluation labels, and missed lesions receive no direct constraint. Better lesion masks can therefore improve the reported PET metrics, while organ-mask quality may affect measured alignment. Label overlap and MTV/TLG change remain proxies for whether registration supports lesion tracking, response assessment, or dose accumulation.

The baseline comparison further shows why deformation regularity must remain a separate endpoint. The selection rule, which weighs accuracy against preservation but not folding, chose a NiftyReg setting that exceeded the folding threshold on 8 of 38 validation and 13 of 38 test pairs. A composite rank can therefore obscure localized failures even when its components include a regularity measure.

5.4 Bone Rigidity

Finite-difference rigidity penalties face a trade-off at bone boundaries: their stencils also constrain surrounding tissue, whereas eroding the masks to avoid this leaves most of the low-resolution skeleton unconstrained. The per-bone rigid fit avoids both, because it compares each bone's displacement with its best rigid motion and needs no derivatives. It produced the lowest within-bone distance distortion among the tested penalties while maintaining soft-tissue alignment. The cost of rigid bones is lower bone Dice, possibly because a rigid bone cannot follow differences between the automatic bone labels of the two sessions as closely as a freely deforming one. Whether this lower overlap reflects worse alignment or an inability of a rigid bone to follow inconsistencies between the automatic bone labels cannot be decided from label overlap alone. The lower within-bone distance distortion shows better preservation of bone shape, but it does not determine whether the bone is placed correctly.

Although about two thirds of the lesion voxels lie on bone labels, adding rigidity on top of the PET losses did not measurably improve lesion volume or uptake preservation in this comparison. Rigidity alone did improve particular PET endpoints in Fig. 2: it reduced network-output $|\Delta\mathrm{MTV}|$ and IO $|\Delta\mathrm{TLG}|$ relative to the corresponding unconstrained variants. Three factors may contribute to the absence of a detected additional benefit on top of the PET losses. First, the PET losses already constrain the predicted lesions. Second, the residual displacement from a rigid fit is still large enough to move boundary voxels of lesions that span only a few voxels. Third, the image-level metrics are dominated by missed lesions, by cancellation between lesions, and by a few patients with large changes (Sec. 4.5). The per-bone rigid fit therefore reduces measured within-bone distance distortion, with a measured cost in bone Dice. The present measures do not establish whether it improves bone placement, and a further PET-preservation benefit may require bones that are rigid to within a fraction of a voxel.

5.5 Limitations

The study uses a single cohort with 28 test patients, so small differences, in particular in MTV and TLG, could not be resolved, and generalization to other scanners and tracers is untested. The images were resampled to about $2.7\times2.7\times3.3$ mm, at which many lesions consist of few voxels and small lesions are dominated by discretization. The organ labels used for evaluation are themselves automatic (TotalSegmentator), and the lesion labels used by IO are predictions with a Dice of about

Table 3 Effect of the PET-preservation and rigidity losses and of where they act on the test set (CT Dice: 28 patients, 38 pairs; metabolic tumor volume [MTV] and total lesion glycolysis [TLG]: 26 patients, 30 pairs); numerical values from Fig. 2. Left: network trained with the losses, without instance optimization (IO); middle: the same network followed by IO with the same losses; right: the network trained without losses followed by IO with the losses. Values are mean±SD over patients after averaging each patient's eligible pairs. †/‡: significantly better/worse than without losses in the same block (for the right block: the network without losses with IO); *: significantly different from the same losses in training and IO (two-sided Wilcoxon signed-rank tests on patient values, Holm-corrected over three comparisons within each column, $p < 0.05$).

	Losses only in training (no IO)			Losses in training and IO			Losses only in IO		
Losses	Dice (%) ↑	|ΔMTV| (%) ↓	|ΔTLG| (%) ↓	Dice (%) ↑	|ΔMTV| (%) ↓	|ΔTLG| (%) ↓	Dice (%) ↑	|ΔMTV| (%) ↓	|ΔTLG| (%) ↓
None	75.5±4.7	10.3±9.5	12.5±11.1	80.2±4.2	11.3±11.9	14.5±14.1	–	–	–
PET	73.6±4.6‡	8.2±9.1†	9.8±10.1	77.7±4.8‡	8.2±10.1	10.3±10.2	78.6±4.4$^{\ddagger *}$	9.3±11.6	11.8±11.7
Rigidity	75.5±4.7	8.9±8.4†	11.0±10.3	79.6±4.2‡	9.2±10.2	10.6±13.4†	79.6±4.2‡	10.0±11.3	11.5±14.2†
PET + rigidity	73.6±4.6‡	8.0±9.0†	9.8±10.9	77.3±4.9‡	7.9±9.7	10.3±10.4	78.3±4.4$^{\ddagger *}$	8.7±10.7	11.3±11.2

0.5, which, as shown above, limits preservation in IO. All patients, including the test patients, were part of the challenge data with which the earlier version of the method was developed; the test patients were not used for any decision in the present study. We studied one network architecture, and we measured the change caused by registration, not its effect on clinical decisions such as response assessment.

6 Conclusion

For similar whole-body PSMA PET/CT data, loss-guided IO is a reasonable choice when predicted masks and per-pair runtime are available and additional anatomical alignment is desired, but the present results do not establish improved PET preservation over network output. Mask coverage, cancellation within image totals, and loss–evaluation mismatch limit the current preservation strategy, so image-level MTV and TLG change should be complemented by per-lesion analysis. The per-bone rigid fit improves measured preservation of internal bone shape, but neither correct bone placement nor additional PET preservation on top of the PET losses was established. Validation on downstream tasks such as lesion tracking and response assessment is needed to determine whether these technical differences are clinically consequential.

7 Appendix A: Effect of the Losses

Table 3 gives the values and significance tests behind Fig. 2.

8 Appendix B: Per-Lesion Uptake Change

9 Appendix C: Tuning of Instance Optimization

Table 4 lists all candidates of the two IO tuning stages on the validation set (Sec. 3.3) and the selected settings.

10 Appendix D: Signed Change of Lesion Volume and Uptake

The main results report the absolute change of MTV and TLG. Table 5 gives the signed change, which shows whether a method systematically enlarges or shrinks the lesions. Every method enlarged the lesions on average (mean ΔMTV $+1.1$ to $+8.2$ %, ΔTLG $+2.8$ to $+9.6$ %), including

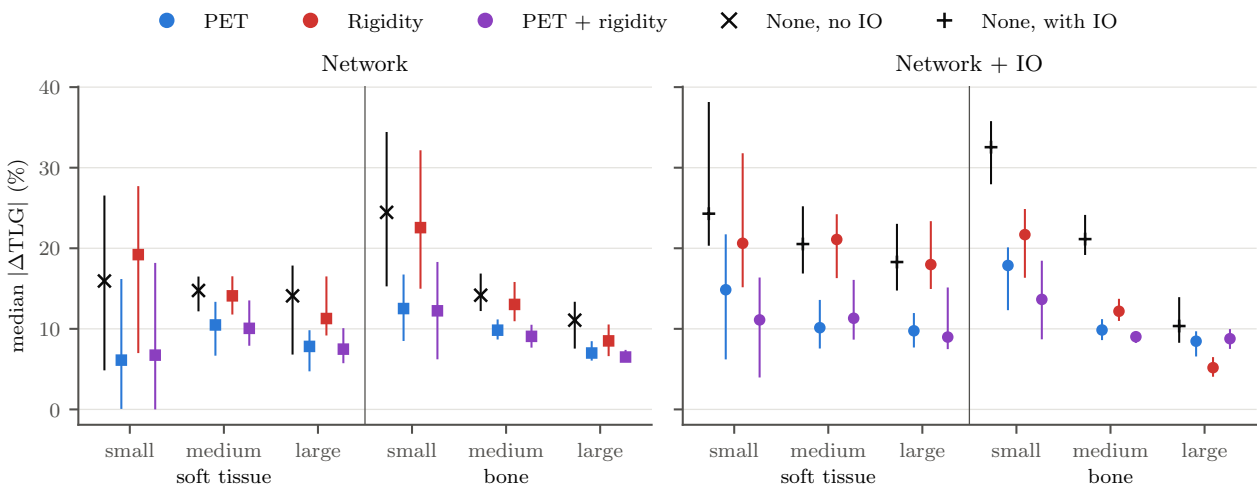


Fig 10 Per-lesion absolute total lesion glycolysis (TLG) change on the test set (1327 lesions from 26 patients and 30 pairs after excluding 108 components smaller than five voxels), stratified by location (soft tissue or bone-associated, with $\geq$50 % of the lesion on bone) and size (small $<$ 0.5 ml, medium 0.5 to $<$ 5 ml, large $\geq$ 5 ml). Points show the median $|\Delta\mathrm{TLG}|$ per lesion; left: network output, right: instance optimization (IO) with the same losses. Within each size stratum, the loss groups are ordered PET, rigidity, and PET plus rigidity; squares and circles denote without and with IO, while $\times$ and $+$ denote the corresponding configurations without losses. Error bars: 95 % confidence intervals from a patient-cluster bootstrap.

the affine pre-registration ($+2.9\,\%$ and $+2.8\,\%$); the spread over patients was larger than these means for every deformable method.

11 Appendix E: Lesion Coverage in Predicted Masks

Tables 6 and 7 give the numbers behind Fig. 5.

12 Appendix F: Rigidity Penalties

Table 8 lists all measures of the rigidity comparison in absolute values, with tests against IO without rigidity, including the three terms of the Staring energy, HD95, and folding in and around the bones.

Table 4 Instance-optimization (IO) tuning on the validation set (CT Dice: 26 patients, 38 pairs; metabolic tumor volume [MTV] and total lesion glycolysis [TLG]: 26 patients, 32 lesion-positive pairs), using the network trained with the PET-preservation and rigidity losses. Stage 1 varies learning rate and iterations; stage 2 varies the common scale of the four PET weights relative to the training-derived values. Within each stage, the chosen setting (bold) has the lowest $|\Delta\mathrm{MTV}|+|\Delta\mathrm{TLG}|$ among candidates within two Dice points of the most accurate one ("Within"). Candidates are sorted by Dice; values are means over patients after averaging eligible pairs; the rigidity weight was fixed at 0.25 (Sec. 3.3).

Candidate	Dice (%)	HD95 (mm)	$\lvert\Delta\mathrm{MTV}\rvert$ (%)	$\lvert\Delta\mathrm{TLG}\rvert$ (%)	Sum (%)	NDV (ppm)	Within
Stage 1: learning rate / iterations							
0.1 / 20	78.7	6.58	10.0	9.8	19.8	15	yes
0.1 / 40	78.6	6.58	10.9	10.6	21.5	13	yes
0.1 / 10	78.1	6.69	7.6	8.4	16.0	17	yes
0.05 / 40	78.0	6.78	6.9	8.1	15.0	29	yes
0.05 / 20	77.8	6.81	6.7	8.0	14.7	29	yes
0.05 / 10	77.0	6.84	5.3	7.0	12.2	34	yes
0.025 / 40	77.0	6.89	4.6	6.2	10.7	12	yes
0.025 / 20	76.7	6.93	4.6	6.3	10.9	12	yes
0.0125 / 40	76.1	6.99	3.9	5.7	9.6	2	no
0.025 / 10	75.7	7.14	4.3	5.5	9.8	18	no
0.0125 / 20	75.6	7.16	3.6	5.1	8.7	2	no
0.0125 / 10	75.3	7.17	3.2	4.6	7.8	3	no
Stage 2: PET weight scale (learning rate 0.025, 40 iterations)							
$\times 0$	79.5	6.61	10.1	11.5	21.6	0	yes
$\times$**0.1**	77.9	6.77	5.6	7.8	13.4	7	yes
$\times 0.25$	77.1	6.86	5.1	6.8	11.9	12	no
$\times 0.5$	77.1	6.87	4.9	6.6	11.5	12	no
$\times 1$	77.1	6.87	5.0	6.7	11.6	12	no
$\times 2$	77.0	6.89	4.9	6.6	11.5	13	no
$\times 4$	77.0	6.89	4.6	6.2	10.8	12	no

Table 5 Signed change in metabolic tumor volume (MTV) and total lesion glycolysis (TLG) caused by registration on the test set (26 patients, 30 lesion-positive pairs): positive values mean that the warped lesions are larger or have greater total uptake. Methods are as in Table 1; Training / IO indicates where losses are used, and PET + rig. denotes PET-preservation and rigidity losses. Values are mean±SD and median over patients after averaging each patient's eligible pairs.

			ΔMTV (%)		ΔTLG (%)	
Method	Training	IO	Mean±SD	Median	Mean±SD	Median
Affine	–	–	$+2.9 \pm 5.8$	$+1.3$	$+2.8 \pm 5.7$	$+1.7$
NiftyReg	–	–	$+4.7 \pm 18.9$	-0.6	$+7.1 \pm 23.8$	-0.1
ConvexAdam	–	–	$+8.0 \pm 8.1$	$+7.1$	$+8.8 \pm 10.3$	$+5.8$
Affine + IO	–	PET + rig.	$+3.5 \pm 5.8$	$+3.8$	$+5.7 \pm 5.6$	$+6.7$
Ours	–	–	$+5.3 \pm 13.1$	$+4.5$	$+7.9 \pm 14.9$	$+7.0$
Ours	PET + rig.	–	$+5.1 \pm 11.0$	$+3.4$	$+7.6 \pm 12.5$	$+5.2$
Ours + IO	–	basic	$+1.1 \pm 16.5$	$+2.1$	$+4.0 \pm 20.0$	$+3.6$
Ours + IO	PET + rig.	PET + rig.	$+4.2 \pm 11.8$	$+3.0$	$+7.3 \pm 12.8$	$+6.4$
Ours + IO	–	PET + rig.	$+3.2 \pm 13.5$	$+3.1$	$+6.2 \pm 14.8$	$+6.2$

Table 6 Metabolic tumor volume (MTV) and total lesion glycolysis (TLG) change of all 1435 provided lesion components in 26 test patients (30 pairs), grouped by how much of each component the predicted lesion mask used by instance optimization (IO) covers; the network is trained without PET losses and IO uses the PET weights scaled by the stated factor. Per image, the change of the summed volume or uptake of the group's lesions relative to the unwarped moving image, in %; mean over the patients with lesions in the group.

	Missed 214 lesions, 6 % of volume		Partly (<50 %) 155 lesions, 6 % of volume		Covered (≥50 %) 1066 lesions, 88 % of volume	
	$\lvert\Delta\text{MTV}\rvert$	$\lvert\Delta\text{TLG}\rvert$	$\lvert\Delta\text{MTV}\rvert$	$\lvert\Delta\text{TLG}\rvert$	$\lvert\Delta\text{MTV}\rvert$	$\lvert\Delta\text{TLG}\rvert$
Network only	12.4	14.5	11.2	11.9	7.9	9.2
+ IO, PET ×0	17.0	18.9	12.0	13.7	9.1	11.0
+ IO, PET ×0.1 (tuned)	14.4	16.5	12.5	13.4	6.0	8.4
+ IO, PET ×1	14.5	16.7	11.5	13.0	5.7	7.7

Table 7 Contribution of all 1435 provided lesion components in 26 test patients (30 pairs), grouped by how much of each component the predicted lesion mask used by instance optimization (IO) covers, to the evaluated metabolic tumor volume (MTV) and total lesion glycolysis (TLG) change; the network is trained without PET losses and IO uses the PET weights scaled by the stated factor. Contribution of a group: the change per image if only the group's lesions changed and all other lesions kept their volume and uptake, in %; mean over patients. All: the evaluation metric. Since changes of different lesions in an image partly cancel, the contributions add up to slightly more.

	Missed 214 lesions, 6 % of volume		Partly (<50 %) 155 lesions, 6 % of volume		Covered (≥50 %) 1066 lesions, 88 % of volume		All lesions 1435 lesions	
	$\lvert\Delta\text{MTV}\rvert$	$\lvert\Delta\text{TLG}\rvert$	$\lvert\Delta\text{MTV}\rvert$	$\lvert\Delta\text{TLG}\rvert$	$\lvert\Delta\text{MTV}\rvert$	$\lvert\Delta\text{TLG}\rvert$	$\lvert\Delta\text{MTV}\rvert$	$\lvert\Delta\text{TLG}\rvert$
Network only	4.6	5.4	1.3	1.0	5.2	6.5	10.3	12.5
+ IO, PET ×0	5.3	6.4	1.3	1.2	5.3	7.3	11.3	14.5
+ IO, PET ×0.1 (tuned)	5.2	6.2	1.4	1.0	3.3	5.4	9.3	11.8
+ IO, PET ×1	5.3	6.3	1.1	0.8	3.3	5.1	9.0	11.5

Table 8 All measures of the rigidity comparison on the test set (alignment and rigidity: 28 patients, 38 pairs; PET preservation: 26 patients, 30 pairs), using the network trained with the PET losses followed by instance optimization (IO) with the PET terms and one rigidity penalty at the weight chosen on the validation set. Rigidity of bones: residual of a rigid fit to each bone (minimized by the rigid-fit penalty), Staring energy (minimized by the Staring penalty), and distortion of short (5–15 mm) and long ($\geq$15 mm) within-bone distances (minimized by neither). HD95: 95th-percentile Hausdorff distance over bones and soft-tissue organs; near-bone soft tissue: soft-tissue organs within three voxels of a bone; Staring energy: sum of its properness, orthonormality, and affinity terms on the whole bone; NDV: non-diffeomorphic volume; NDV in bone band: in a three-voxel band around the bones. Mean$\pm$SD over patients; †/‡: significantly better/worse than without rigidity (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected over the three penalties per row, $p < 0.05$).

Measure	No rigidity	Staring ($\times$3)	Staring + erosion ($\times$100)	Per-bone rigid fit ($\times$100)
Alignment				
CT Dice (%)	77.7±4.8	75.7±4.3‡	77.6±4.6	76.1±4.6‡
CT HD95 (mm)	6.96±2.35	7.03±2.35‡	6.90±2.34	7.03±2.35
Bone Dice (%)	78.4±6.4	74.6±5.7‡	77.5±6.2‡	74.5±6.2‡
Bone HD95 (mm)	7.53±3.38	7.68±3.40‡	7.54±3.36	7.79±3.41‡
Soft-tissue Dice (%)	79.9±3.9	79.8±3.9	80.7±3.9†	80.8±4.0†
Soft-tissue HD95 (mm)	6.31±2.94	6.29±2.95	6.18±2.98†	6.18±3.01†
Near-bone soft-tissue Dice (%)	74.1±4.3	73.7±4.2‡	74.9±4.2†	74.8±4.2
Rigidity of bones				
Rigid-fit residual (mm)	2.67±0.52	2.00±0.48†	2.31±0.47†	1.39±0.40†
Staring energy	3.517±2.110	0.272±0.090†	1.803±0.650†	1.419±0.473†
Staring properness	0.182±0.100	0.022±0.009†	0.097±0.033†	0.093±0.027†
Staring orthonormality	1.431±0.840	0.160±0.066†	0.765±0.289†	0.733±0.261†
Staring affinity	1.903±1.180	0.090±0.040†	0.942±0.339†	0.594±0.205†
Distortion short (%)	13.41±3.62	5.53±1.04†	8.76±1.04†	5.01±1.13†
Distortion long (%)	4.44±0.83	3.17±0.68†	3.53±0.63†	1.83±0.62†
Regularity				
NDV (ppm)	7.9±29.2	7.9±29.3†	0.7±0.6	0.6±0.6†
NDV in bone (ppm)	6.5±4.6	0.3±1.2†	16.5±23.6‡	9.6±23.7
NDV in bone band (ppm)	1.9±1.7	0.7±0.9†	3.7±4.0‡	3.6±4.6‡
Preservation				
$\lvert\Delta\mathrm{MTV}\rvert$ (%)	8.2±10.1	7.1±8.7	7.9±10.0	8.0±8.9
$\lvert\Delta\mathrm{TLG}\rvert$ (%)	10.3±10.2	9.2±9.4	9.6±10.2	10.5±9.4

13 Appendix G: Implementation and Tuning Details

Loss weights. Table 9 lists the weights used in network training and IO.

Table 9 Loss weights of Eq. (3) in network training (full-resolution level; the Dice weight is 3, 4, and 5 at the three levels) and in instance optimization (IO).

	λ_{NCC}	λ_{DSC}	λ_{MTV}	$\lambda_{\overline{J}}$	λ_{J}	λ_{TLG}	λ_{rig}	$\lambda_{\mathrm{smooth}}$	λ_{NDV}
Training	5	5	5	0.125	1.25	1.25	0.05	10	10^4
IO	5	5	$4 \cdot 10^4$	10^3	4	16	0.25	0	10

Network training. The quarter-, half-, and full-resolution levels were trained with Adam[22] for 10^5, 10^5, and $1.2 \cdot 10^5$ steps, respectively, with learning rates of $1.5 \cdot 10^{-4}$, $1.2 \cdot 10^{-3}$, and 10^{-5}. We used a batch size of one with gradients accumulated over four pairs, a linear warm-up over five epochs, and a cosine decay to 10 % of the initial rate. A finer level was trained jointly with the coarser ones after ten epochs. Convolutions ran in bfloat16 and all losses in single precision. Training took 18 h, 22 h, and 3.7 days for the three levels on one NVIDIA A100 (80 GB). The intensity augmentations were CT shifts of ± 0.02 in the normalized window (about ± 50 HU), CT scaling between 0.9 and 1.1, and PET scaling between 0.85 and 1.15. Spatial augmentations were left–right flipping ($p = 0.5$) and field-of-view mismatch produced by cropping up to 40 axial slices from both images and up to 10 further slices from each image separately.

Challenge ranking. For each metric, all 72 directed pairs among the nine resampled methods were compared with one-sided Wilcoxon tests and Holm correction, and a method scored (number of methods it significantly outperformed $+1$)/K for K methods. Scores were averaged into accuracy (CT Dice, HD95), PET preservation ($|\Delta\mathrm{MTV}|$, $|\Delta\mathrm{TLG}|$), and regularity (NDV, after subtracting a practical-equivalence threshold of 0.005 % per pair). The composite score was $100 \cdot \mathrm{acc}^{0.4}\,\mathrm{PET}^{0.4}\,\mathrm{reg}^{0.2}$. The challenge additionally scores Dice and HD95 of organs with physiological PET uptake, for which no labels are available to us, so we omitted them. We also omitted lesion Dice and HD95 because lesions appear, disappear, and change between sessions. The 95 % rank intervals use 1000 patient bootstrap samples.

Tuning grids. IO was tuned step-wise on the network trained with both losses: first learning rate $\times$ iterations ($\{0.0125, 0.025, 0.05, 0.1\} \times \{10, 20, 40\}$), and then a common scale of the four PET weights ($\times\{0, 0.1, 0.25, 0.5, 1, 2, 4\}$ of the training-derived values). The selected setting was a learning rate of 0.025 with 40 iterations and a PET factor of $\times 0.1$ (Table 4). NiftyReg[31,32] used `reg_aladin` followed by GPU `reg_f3d` with normalized mutual information, because local NCC is unavailable on the GPU. Its grid combined bending-energy weights 0.0002, 0.0005, 0.001, 0.005, and 0.02 with final control-point spacings of 3, 5, 7, and 9 voxels; the selected setting used 0.0002 and 7 voxels. ConvexAdam[19] used its own ANTs affine registration with a $[-300, 300]$ HU CT window and regularization weights $\lambda \in \{0.1, 0.25, 0.5, 1.25, 2, 4\}$; the selected setting used 2. Several selected settings lay at the edge of their grid, but extending the grids changed the validation scores only marginally. The post-hoc accuracy–preservation sweep used PET-weight

factors $\{0, 10^{-4}, 3{\cdot}10^{-4}, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1\}$ relative to the training-derived values. The per-lesion terms were tested at $\{1, 10, 100\}$ times base weights of 8 (volume), 8 (uptake), and 15 (mean Jacobian determinant), i.e. 0.1 times their initial values, like the other PET weights; the validation set selected $\times 10$. For the rigidity comparison, the Staring and eroded-Staring reference weights were 0.66 and 1.03, respectively, after calibration against the rigid-fit weight of 0.25. All three penalties were evaluated at $\{0.1, 0.3, 1, 3, 10, 30, 100\}$ times their reference weight; the selected factors were $\times 3$ for Staring and $\times 100$ for eroded Staring and the rigid fit.

Disclosures

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper.

Code, Data, and Materials Availability

All experiments use the publicly available training data of the Learn2Reg 2026 PSMAReg challenge,[3,4] which is adapted from the PSMA-PET-CT-Lesions dataset in The Cancer Imaging Archive[5] (DOI: 10.7937/r7ep-3x37). The private test data of the challenge were not used. The code, the patient-level split into training, validation, and test sets, and the trained model weights are available at https://github.com/koegl/psmareg-extended under the MIT license.

Acknowledgments

F.K. and J.A.S. are partially supported by the German Federal Ministry of Research, Technology and Space (DECIPHER-M, 01KD2420G). A.R. is supported by funding from the Free State of Bavaria (StMGP) as part of the project "KI-gestützte multi-modale Diagnostik und stratifizierte Therapie für Endometriose (EndoKI)". D.M.L. and J.A.S. received funding from Helmholtz Imaging, a platform of the Helmholtz Information & Data Science Incubator. L.F. was partially funded by the Intramural Research Funding Grant "AI-driven Longitudinal Lesion Tracking" of the Faculty of Medicine, University of Augsburg, and by the Bavarian Center for Cancer Research as part of the Lighthouse "Local Therapies". Additional funding came from the Bavarian Ministry of Economic Affairs, Regional Development and Energy (StMWi) under grant number DIK-2310-0004//DIK0556/02.

References

1 Wolfgang P. Fendler, Matthias Eiber, Mohsen Beheshti ... Ken Herrmann, "68Ga-PSMA PET/CT: Joint EANM and SNMMI procedure guideline for prostate cancer imaging: version 1.0," *European Journal of Nuclear Medicine and Molecular Imaging* **44**, 1014–1024 (2017).

2 Learn2Reg Consortium, "Learn2Reg 2026 – PSMAReg: Registration of pre-therapy and follow-up whole-body PSMA PET/CT scans." Codabench challenge (2026). https://www.codabench.org/competitions/15724/.

3 Lasse Hansen, Wiebke Heyer, Christoph Großbröhmer ... Mattias P. Heinrich, "Learn2reg 2024: New benchmark datasets driving progress on new challenges," *Machine Learning for Biomedical Imaging* **3**, 775–791 (2025).

4 Alessa Hering, Lasse Hansen, Tony C. W. Mok . . . Mattias P. Heinrich, "Learn2reg: Comprehensive multi-task medical image registration challenge, dataset and evaluation in the era of deep learning," *IEEE Transactions on Medical Imaging* **42**(3), 697–712 (2023).

5 Katharina Jeblick, Stefan O. Schönberg, and others, "A whole-body PSMA-PET/CT dataset with manually annotated tumor lesions (PSMA-PET-CT-Lesions)." The Cancer Imaging Archive (2024). Version 1.

6 Tom Vercauteren, Xavier Pennec, Aymeric Perchant . . . Nicholas Ayache, "Diffeomorphic demons: Efficient non-parametric image registration," *NeuroImage* **45**(1, Supplement 1), S61–S72 (2009).

7 B.B. Avants, C.L. Epstein, M. Grossman . . . J.C. Gee, "Symmetric diffeomorphic image registration with cross-correlation: Evaluating automated labeling of elderly and neurodegenerative brain," *Medical Image Analysis* **12**(1), 26–41 (2008). Special Issue on The Third International Workshop on Biomedical Image Registration - WBIR 2006.

8 D. Rueckert, L.I. Sonoda, C. Hayes . . . D.J. Hawkes, "Nonrigid registration using free-form deformations: application to breast mr images," *IEEE Transactions on Medical Imaging* **18**(8), 712–721 (1999).

9 Guha Balakrishnan, Amy Zhao, Mert R. Sabuncu . . . Adrian V. Dalca, "An unsupervised learning model for deformable medical image registration," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 9252–9260 (2018).

10 Guha Balakrishnan, Amy Zhao, Mert R. Sabuncu . . . Adrian V. Dalca, "VoxelMorph: A learning framework for deformable medical image registration," *IEEE Transactions on Medical Imaging* **38**(8), 1788–1800 (2019).

11 Adrian V. Dalca, Guha Balakrishnan, John Guttag . . . Mert R. Sabuncu, "Unsupervised learning for fast probabilistic diffeomorphic registration," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018*, Alejandro F. Frangi, Julia A. Schnabel, Christos Davatzikos . . . Gabor Fichtinger, Eds., 729–738, Springer International Publishing, (Cham) (2018).

12 Junyu Chen, Eric C. Frey, Yufan He . . . Yong Du, "Transmorph: Transformer for unsupervised medical image registration," *Medical Image Analysis* **82**, 102615 (2022).

13 Tony C. W. Mok and Albert C. S. Chung, "Large deformation diffeomorphic image registration with laplacian pyramid networks," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2020*, Anne L. Martel, Purang Abolmaesumi, Danail Stoyanov . . . Leo Joskowicz, Eds., 211–221, Springer International Publishing, (Cham) (2020).

14 Fryderyk Kögl, Anna Reithmeir, Daniel M. Lang . . . Julia A. Schnabel, "PET quantification-preserving registration for longitudinal whole-body PSMA PET/CT," (2026). arXiv:2610.21451.

15 Jakob Wasserthal, Hanns-Christian Breit, Manfred T. Meyer . . . Martin Segeroth, "Totalsegmentator: Robust segmentation of 104 anatomic structures in CT images," *Radiology: Artificial Intelligence* **5**(5), e230024 (2023).

16 Katharina Jeblick, Balthasar Schachtner, Andreas Mittermeier . . . others, "A whole-body PSMA-PET/CT dataset with manually annotated tumor lesions," *Scientific Data* **13**, 1023 (2026).

17 Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl . . . Klaus H. Maier-Hein, "nnu-net: a self-configuring method for deep learning-based biomedical image segmentation," *Nature Methods* **18**, 203–211 (2021).

18 Maximilian Rokuss, Balint Kovacs, Yannick Kirchhoff . . . Fabian Isensee, "From fdg to psma: A hitchhiker's guide to multitracer, multicenter lesion segmentation in pet/ct imaging." arXiv:2409.09478 (2024).

19 Hanna Siebert, Lasse Hansen, and Mattias P. Heinrich, "Fast 3d registration with accurate optimisation and little learning for learn2reg 2021," in *Biomedical Image Registration, Domain Generalisation and Out-of-Distribution Analysis*, Marc Aubreville, David Zimmerer, and Mattias Heinrich, Eds., 174–179, Springer International Publishing, (Cham) (2022).

20 Brian B. Avants, Nicholas J. Tustison, Gang Song . . . James C. Gee, "A reproducible evaluation of ants similarity metric performance in brain image registration," *NeuroImage* **54**(3), 2033–2044 (2011).

21 Vincent Arsigny, Olivier Commowick, Xavier Pennec . . . Nicholas Ayache, "A log-Euclidean framework for statistics on diffeomorphisms," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, *Lecture Notes in Computer Science* **4190**, 924–931, Springer (2006).

22 Diederik P. Kingma and Jimmy Ba, "Adam: A method for stochastic optimization," in *International Conference on Learning Representations (ICLR)*, (2015). arXiv:1412.6980.

23 Wolfgang Kabsch, "A solution for the best rotation to relate two sets of vectors," *Acta Crystallographica Section A* **32**(5), 922–923 (1976).

24 S. Umeyama, "Least-squares estimation of transformation parameters between two point patterns," *IEEE Transactions on Pattern Analysis and Machine Intelligence* **13**(4), 376–380 (1991).

25 Yihao Liu, Junyu Chen, Shuwen Wei . . . Jerry Prince, "On finite difference jacobian computation in deformable image registration," *International Journal of Computer Vision* **132**, 3678–3688 (2024).

26 Marius Staring, Stefan Klein, and Josien P. W. Pluim, "A rigidity penalty term for nonrigid registration," *Medical Physics* **34**(11), 4098–4108 (2007).

27 Frank Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin* **1**(6), 80–83 (1945).

28 Sture Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics* **6**(2), 65–70 (1979).

29 Bradley Efron and Robert J. Tibshirani, *An Introduction to the Bootstrap*, Chapman & Hall, New York (1993).

30 Neel Dey and Adrian V. Dalca, "Registration is a representation problem: Solving whole-body pet/ct registration," (2026).

31 Marc Modat, Gerard R. Ridgway, Zeike A. Taylor . . . Sébastien Ourselin, "Fast free-form deformation using graphics processing units," *Computer Methods and Programs in Biomedicine* **98**(3), 278–284 (2010). HP-MICCAI 2008.

32 Marc Modat, David M. Cash, Pankaj Daga . . . Sébastien Ourselin, "Global image registration using a symmetric block-matching approach," *Journal of Medical Imaging* **1**(2), 024003 (2014).

Biographies

Fryderyk Kögl is a doctoral researcher at the Technical University of Munich and Helmholtz Munich. He received his BSc degree in engineering science and MSc degree in biomedical computing from the Technical University of Munich. His research focuses on learning-based medical image registration, data curation and visualization, and neuronavigation.

Anna Reithmeir is a doctoral researcher at the Technical University of Munich and the Munich Center for Machine Learning. She received her BSc and MSc degrees in informatics from the Technical University of Munich. Her research focuses on longitudinal and multimodal medical image registration, physically plausible deformation models, and machine-learning methods for abdominal imaging.

Lukas Förner is a doctoral researcher in Clinical Computational Medical Imaging at University Hospital Augsburg. He holds an MSc degree and works at the interface of medical imaging and artificial intelligence. His research focuses on machine-learning methods for multimodal and longitudinal imaging, including image registration and quantitative PET/MRI analysis.

Daniel M. Lang is a postdoctoral research scientist at Helmholtz Munich. He received his doctoral degree from the Technical University of Munich for research on deep-learning analysis of oncological imaging data. His research focuses on dynamic medical imaging, generative modeling, self-supervised learning, and image analysis in oncology.

Emily M. Chan is a postdoctoral research scientist at Helmholtz Munich. She received her MEng degree in biomedical engineering from Imperial College London and her PhD degree from King's College London. Her research focuses on medical image computing, transfer learning, multimodal learning, and machine-learning methods for clinically relevant radiological tasks.

Rickmer Braren is director of the Department of Diagnostic and Interventional Radiology and Nuclear Medicine at University Medical Center Hamburg-Eppendorf. He studied medicine and received his doctoral degree at University Medical Center Hamburg-Eppendorf before completing his habilitation at the Technical University of Munich. His research focuses on quantitative oncological imaging and explainable artificial intelligence for disease detection and prediction.

Julia A. Schnabel is director of the Institute of Machine Learning in Biomedical Imaging at Helmholtz Munich, professor at the Technical University of Munich, and chair in Computational Imaging at King's College London. She received her PhD degree in computer science from University College London and is a Fellow of IEEE, ELLIS, and the MICCAI Society. Her research focuses on machine learning for medical imaging, motion modeling, and multimodal and quantitative imaging.